

Racing Against Automation: Dynamic Incentives When AI Can Preempt the Human *

Feng Tian

HKU Business School, The University of Hong Kong, fengtian@hku.hk

Yangge Xiao

Faculty of Business and Economics, The University of Melbourne, yangge.xiao@unimelb.edu.au

Yufei Zhu

Institute of Operations Research and Analytics, National University of Singapore, yufei_zhu@u.nus.edu

Ying-Ju Chen

The Hong Kong University of Science and Technology, imchen@ust.hk

Recent advances in artificial intelligence (AI), and AI agents in particular, are reshaping innovation by enabling autonomous search, evaluation, and iteration toward new discoveries. Yet human researchers remain essential in novel and poorly structured settings. A human agent may be more productive but privately chooses effort, creating a moral hazard problem; the AI agent requires no incentives but is costly to operate and can preempt the human agent's success. This raises a natural question: When, if ever, should the firm race the AI agent against the human agent rather than rely on either one alone? We develop a continuous-time principal-agent model in which a principal draws on both a human agent and an independent AI agent in the search for innovation. We establish that when the AI agent is not cost-effective, the principal never deploys it and relies on human-alone search. In contrast, when the AI agent is cost-effective, a natural conjecture is that the principal should always keep it active as a rival, since it can accelerate breakthroughs and, by threatening to preempt the human agent, strengthen the human agent's incentives. We show that this intuition can fail: persistent AI rivalry may lower both the human agent's and the principal's payoffs relative to human-alone search, so aggressive competition can hurt both parties. Instead, the optimal contract may begin with human-alone search, transition to a human-AI race, and finally switch to autonomous AI search if no breakthrough occurs. Relative to a human-human benchmark, deploying the AI agent alongside the human can also align the preferences of the principal and the focal human agent over whether competition should come from an AI agent or another human. In an extension with imperfect AI-generated outcomes, the optimal contract features time-varying verification. In sum, AI rivalry is a double-edged sword: it can accelerate discovery and discipline human effort, but deploying it too early crowds out valuable human-alone search, so firms may optimally delay it even when the AI is productive.

Key words: Dynamic Contracting, Moral Hazard, Human-AI Race, Innovation Discovery, Preemption

* We thank the seminar participants at HKU Summer Workshop (June 10, 2026) for useful feedback.

1. Introduction

Innovation is a primary driver of firm value and a central source of competitive advantage in dynamic markets. Firms therefore invest substantial resources in sourcing and organizing innovative activity. Traditionally, such activity has relied on human ingenuity. Recent advances in artificial intelligence (AI), however, are beginning to reshape how innovation is produced. The first wave of AI adoption centered on predictive and generative models that served as *tools* for human researchers, augmenting specific tasks in the innovation process such as prediction, simulation, etc. More recently, AI *agents* have begun to move beyond this tool-based role. Rather than serving merely as tools that researchers wield, these systems can plan, execute, evaluate, and iterate through parts of the innovation process with minimal human intervention. This shift raises the possibility that AI may no longer serve only as a tool in human research, but may instead become an independent channel of discovery.

Several recent advances illustrate this possibility. Google DeepMind’s AlphaTensor discovered matrix-multiplication algorithms more efficient than any humans had devised in over fifty years (Fawzi et al. 2022), while AlphaDev found faster sorting routines later adopted into the standard C++ library, the first such change in more than a decade (Mankowitz et al. 2023). In both cases, the AI agent carried out the discovery process itself: it searched over candidate solutions, evaluated them against explicit performance objectives, and selected those that improved on established benchmarks. Similar patterns are emerging beyond algorithm design. In the life sciences, AI agents have designed nanobodies later validated experimentally (Swanson et al. 2025), and agentic systems are increasingly being developed for use across the drug-discovery pipeline (Seal et al. 2025). The shift is also entering managerial practice: in a 2025 survey of 2,000 CEOs by the IBM Institute for Business Value, 61% reported that their organizations were actively adopting AI agents and preparing to deploy them at scale (IBM Institute for Business Value 2025).

Human researchers, however, remain indispensable in genuinely novel environments, where data are sparse, feedback is limited, and the problem itself is not yet well structured. In such settings, innovation requires not only computation but also judgment: identifying promising directions, forming hypotheses from limited evidence, and exercising forms of creative reasoning that cannot be fully inferred from historical data (Broekhuizen et al. 2023, Prasad et al. 2025). These are also settings in which current AI agents face important limits (Bhardwaj 2025). Human researchers therefore remain more productive than AI agents on poorly structured innovation tasks.

At the same time, relying on human researchers creates an incentive problem. A firm cannot directly observe whether a human researcher is exerting effort or shirking, and therefore faces a moral hazard problem (Holmström 1979, Sannikov 2008). By contrast, an autonomous AI agent does not create the same incentive friction because its operation is directly controlled by the firm

(Hadfield and Koh 2025). Human researchers may therefore have a productivity advantage in environments that require judgment under uncertainty, whereas AI agents do not require incentive provision but incur operating costs once deployed.

For a firm seeking innovation, these forces create a central deployment tradeoff. The firm can rely on the human agent alone, deploy the AI agent alone, or race the two in parallel toward the same breakthrough. Racing increases the likelihood of innovation because either agent may succeed. At the same time, it introduces a preemption effect: an AI-generated breakthrough can arrive before the human agent’s breakthrough, altering the human agent’s incentives to exert effort. This raises our central question: *When, if ever, should the firm race the AI agent against the human agent rather than rely on either one alone?*

To answer this question, we develop a continuous-time dynamic principal–agent model in which a principal has access to two productive channels, a human agent and an AI agent, to generate an innovation breakthrough. Success arrives stochastically according to a Poisson process, with the instantaneous arrival rate determined by the active channels. The human agent *privately* chooses whether to exert costly effort, creating a moral hazard problem, while the principal *publicly* chooses whether to deploy the AI agent. If both agents are active, their breakthrough rates add up. Activating either channel requires paying the corresponding operating cost. Once a breakthrough occurs, the principal observes whether it was generated by the human agent or the AI agent, collects a fixed reward, makes a contractual transfer to the human agent, and the game ends. The principal designs an optimal contract specifying when to activate each agent and how to compensate the human agent upon a breakthrough.

As a preliminary step, we show that when the AI agent is cost-ineffective, the principal never deploys it, and relies on the human alone throughout. In contrast, when the AI agent becomes cost-effective, a natural conjecture is that the principal should always keep it active as a rival to the human agent. After all, the AI agent increases the arrival rate of breakthroughs and, when the human agent is also active, can reduce the rent required to motivate him by possibly preempting his success. We show that this conjecture does not generally hold. Persistent AI agent rivalry can make both the principal and the human agent worse off relative to the human-alone search. The key tension is between two opposing effects of AI preemption: an incentive effect, whereby rivalry from the AI agent disciplines the human agent and reduces the rent the principal must pay; and an efficiency effect, whereby preemption by the AI agent crowds out the potentially valuable human channel and destroys the surplus it would have generated. When the AI agent is only weakly effective, the efficiency loss dominates the incentive saving, revealing that overly intense competition can harm both the principal and the human agent.

We then characterize the optimal contract when the AI agent is cost-effective. The principal may initially let the human agent work alone, then introduce the AI agent as a rival, and eventually rely on the AI agent alone. The logic behind these transitions reflect a tradeoff: the contract introduces competition from the AI agent only when its incentive benefits outweigh the efficiency loss from preempting the human channel. Once the human channel is no longer worth maintaining, the contract switches to the AI agent alone. Thus, the problem is not AI agent rivalry per se, but excessive rivalry. Furthermore, as the AI agent becomes more effective, the efficiency loss from preemption shrinks and the incentive benefits increase, so the principal introduces AI agent rivalry earlier. In particular, when the AI agent becomes moderately effective, the human-alone stage disappears and the AI agent is deployed from the outset; when the AI agent is sufficiently effective, the principal bypasses the human agent entirely and uses the AI agent alone.

Building on this characterization, we compare the AI agent with an alternative source of competition: a second human agent. A second human agent provides an additional discovery channel but also creates a moral hazard problem. The principal and the focal human agent may have conflicting preferences over whether competition should come from the AI agent or from another human agent. When the AI agent is sufficiently weak, the principal benefits more from a human–human race, while the focal human agent prefers AI agent rivalry. When the AI agent is sufficiently strong, the opposite holds. For an intermediate AI agent, both parties agree, though their aligned preference depends on the AI agent’s flow cost: when the flow cost of deploying the AI agent is sufficiently low, both prefer the human–AI rivalry; when the flow cost is sufficiently high, both prefer the human–human race.

Finally, we extend the model to imperfect AI-generated outcomes. In practice, AI agents may behave inconsistently or produce plausible but incorrect outputs that are difficult to detect (Chen et al. 2025, Rabanser et al. 2026), imposing a nontrivial verification burden on the firm (Passi 2025). In our extension, we allow outcomes generated by the AI agent to be fake and let the principal decide whether to inspect them. Imperfect AI-generated outcomes add a new reason to delay deployment of the AI agent: early deployment may terminate the project not only through a genuine breakthrough, but also through false termination if a fake output is accepted. Once the AI agent is deployed, the optimal contract may initially inspect AI-generated outcomes while the continuation value of search is high, and later stop inspecting when this value declines. Imperfect AI-generated outcomes do not overturn the delayed-racing pattern of the base model; but they refine the racing phase into inspected and uninspected AI agent competition.

These results yield several important managerial implications. First, AI agent rivalry can be a double-edged sword. It can accelerate discovery and discipline human innovators, but introducing it too early may crowd out valuable human-alone search. Firms may optimally delay competition

from the AI agent even when the AI agent is productive on its own. Second, the choice between deploying the AI agent and hiring another human agent depends on the AI agent’s capability and cost. A low-cost, moderately capable AI agent can be attractive to both the firm and the incumbent human agent, while a highly capable AI agent benefits the firm but harms the human agent by intensifying competition. Finally, when AI-generated outputs are imperfect, firms should coordinate deployment and verification policies: inspection is most valuable early, when rejecting a false AI-generated output preserves a valuable continuation option, and becomes less valuable as time goes by.

The rest of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 introduces the model. Section 4 establishes benchmarks and characterizes the optimal contract in the base model. Section 5 compares human–AI and human–human organizations. Section 6 extends the model to imperfect AI-generated outcomes and costly inspection. Section 7 concludes.

2. Literature Review

Our work sits at the intersection of four streams of literature: dynamic contracting under moral hazard, incentive design with AI, innovation race and contest, and costly verification.

Dynamic contracting and moral hazard. Our paper contributes to the literature on contracting under moral hazard. Following the classical static formulation of Holmström (1979), a large literature studies how principals provide incentives when effort is hidden. The problem was subsequently extended to various settings, such as dynamic contracting (Spear and Srivastava 1987), agents with limited liability (Wang and Hu 2021), delegated search (Zorc et al. 2025), robust contracts (Carroll 2015).

Our model is closest to continuous-time dynamic moral-hazard models. Sannikov (2008) provides a foundational continuous-time principal–agent framework in which the agent’s effort affects the drift of a Brownian motion. Another strand studies environments in which effort affects the arrival rate of payoff-relevant Poisson events, including Biais et al. (2010), Myerson (2015), Green and Taylor (2016), Chen et al. (2020), Sun and Tian (2018), and Tian et al. (2026). We depart from this literature by allowing the principal to dynamically choose which productive channels to activate: a strategic human agent, a non-strategic autonomous AI agent, both, or neither.

This also connects our paper to dynamic contracting with multiple productive agents, including Che and Yoo (2001), Georgiadis (2015), Shan (2017), and Tian et al. (2026). Among these papers, the closest to ours is Tian et al. (2026), who study dynamic contracting with multiple strategic agents working toward a breakthrough. Our model differs in two key respects. First, in our model the active channels generate independent and additive breakthrough hazards, whereas Tian et al. (2026) study an environment in which agents’ efforts are substitutable. Second, our second productive channel is an autonomous AI agent rather than another strategic human agent. Thus, the

principal gains an additional source of breakthroughs without facing a moral-hazard problem from the second productive channel, an autonomous AI agent. These differences lead to fundamentally different optimal contracts. We further compare the human–AI organization with a human–human organization to isolate how the nature of the second productive channel affects both the principal and the incumbent human agent.

Incentive design with AI. With the rise of artificial intelligence, human–AI interaction has attracted growing attention; see [Castro et al. \(2023\)](#), [Chen et al. \(2024\)](#), [Boyacı et al. \(2024\)](#), [De Véricourt and Gurkan \(2026\)](#), [Aouad et al. \(2026\)](#), [Baek et al. \(2026\)](#) and also [Cohen et al. \(2026\)](#) for a recent review. Within this broad area, a growing literature studies how principals design incentives for human agents in the presence of AI. One strand of this literature focuses on information provision as an incentive device in human–AI systems, including algorithmic recommendations, and AI-generated signals; see [Boyacı et al. \(2024\)](#), [McLaughlin and Spiess \(2024\)](#), [Guan et al. \(2025\)](#).

Our work complements this stream by studying monetary incentives. In [Zhang et al. \(2025\)](#), a platform uses a revenue-sharing mechanism to motivate a representative content creator who can delegate content production to AI. [Dai and Taylor \(2025\)](#) study a setting in which the principal jointly chooses the design of the AI system, captured by its temperature, and the payment scheme for a human agent whose output depends on both human effort and the AI system. [Bastani and Cachon \(2025\)](#) study output-based contracts when AI may produce an incorrect output and a human agent can exert effort to check and improve it. [Siderius et al. \(2026\)](#) extend this framework to a dynamic setting in which AI also affects the evolution of human expertise. [Mao et al. \(2026\)](#) study optimal medical liability design in a principal–agent setting where a social planner uses liability to regulate a physician with privately observed diagnostic quality. The physician chooses among standard care, a personalized judgment-based treatment, and following an imperfect AI recommendation; liability is imposed when the physician deviates from standard care and the treatment subsequently fails. In an independent and concurrent work, [Abdolmaleki and Duenyas \(2026\)](#) also consider a dynamic contracting problem in which the principal decides when to adopt AI. In their model, however, AI is a tool adopted by the human agent to assist production, so output generated with AI is attributed to the human agent. In summary, these papers study incentive design in human–AI collaboration, where AI assists, substitutes for, or changes the human agent’s production task.

Our paper departs from these studies by treating AI as an independent agent ([Immorlica et al. 2024](#), [Hadfield and Koh 2025](#), [Cheng et al. 2025](#)). We study a different organizational role for AI: AI as a rival in a race for a breakthrough. In our setting, the AI agent is an independent discovery channel that can generate a breakthrough before the human agent and thereby preempt him. This

changes the incentive problem in a fundamental way. In collaborative settings, a more capable AI system may hurt the principal by making it harder to incentivize the human agent: as AI assistance becomes more effective, the human agent can rely more on AI tools and has weaker incentives to exert costly effort. In our racing setting, by contrast, a more capable AI agent always benefits the principal: it reduces the rent required to incentivize the human agent and lowers efficiency losses by crowding out the less valuable human channel. Our paper highlights how AI competition, rather than AI collaboration, reshapes dynamic incentive provision.

Innovation race and contest. Our human–AI competition framework connects to the innovation race literature. [Loury \(1979\)](#) develops the canonical innovation race model in which competing firms invest in R&D and innovation arrives stochastically. [Reinganum \(1981, 1982\)](#) extends this framework using dynamic and differential games to model the competition, and [Grossman and Shapiro \(1987\)](#) and [Harris and Vickers \(1985, 1987\)](#) study how competition evolves as firms gain leads or fall behind in an innovation race. A common feature of innovation race models is that the first breakthrough preempts rivals from obtaining the innovation prize ([Loury 1979](#), [Reinganum 1981, 1982](#)). This preemption effect is also central to our model, where the possibility of AI preempting the human agent reshapes the agent’s incentive to exert effort. We depart from this literature in two respects. First, the race in our model is between a strategic human agent and a non-strategic, autonomous AI, rather than between strategic rival firms. Second, we study *delegated* innovation: a principal decides when and how to deploy each agent and designs the incentive scheme.

The delegation of innovation to multiple agents also connects our model to the innovation contest literature, in which a principal delegates an innovation project to contestants who bear their own R&D costs and compete for a prize. The classical contest literature studies how to optimally design prize structures, entry rules, and award schemes to maximize innovation quality; these models are largely static, with the principal committing ex ante to a fixed contest rule ([Taylor 1995](#), [Terwiesch and Xu 2008](#), [Ales et al. 2017](#)). A more recent strand examines optimal feedback and disclosure design in dynamic contests ([Halac et al. 2017](#), [Bimpikis et al. 2019](#), [Mihm and Schlapp 2019](#), [Chen et al. 2022](#), [Ely et al. 2023](#)). In contrast to both strands, we study a dynamic contracting environment in which the principal directly bears agent deployment costs and designs a continuous-time incentive scheme, jointly optimizing payment and deployment over the course of the project.

Inspection. Our inspection extension is also related to the literature on costly verification. One stream studies adverse selection problems, in which a principal can choose whether to verify agents’ reports; see [Townsend \(1979\)](#), [Ben-Porath et al. \(2014\)](#), [Mylovanov and Zapechelnjuk \(2017\)](#), [Li and Libgober \(2026\)](#). Another stream studies moral hazard problems, in which a principal can inspect whether agents have shirked; see [Chen et al. \(2020\)](#), [Kim and Xu \(2024\)](#), [Ball and Knoepfle \(2023\)](#). Our work differs from both streams by studying verification of the output of a non-strategic,

non-shirking agent. In particular, we examine a human–AI contracting environment in which AI-generated breakthroughs may be false, and the principal decides whether to incur a verification cost or accept the AI output and terminate the project.

3. Model

We consider a principal–agent model in which a principal works with two productive agents, a strategic human agent and a non-strategic AI agent, to generate an innovation breakthrough. The breakthrough arrives randomly according to a Poisson process whose rate is jointly determined by the AI agent’s activity and the human agent’s effort. In particular, at each time $t \geq 0$, if no breakthrough has yet occurred, the principal chooses whether to activate the human agent and the AI agent. Let $\nu_h(t) \in \{0, 1\}$ and $\nu_a(t) \in \{0, 1\}$ denote the principal’s recommended effort for the human agent and designated activity for the AI agent, respectively. Here, 1 means that an agent is activated and 0 means that the agent is not activated. The AI agent is non-strategic, so its actual activity coincides with the principal’s designated process $\nu_a(\cdot)$, which is publicly observable. In other words, the AI agent creates no moral hazard problem. By contrast, when the human agent is recommended to work, he can privately choose his actual effort $\tilde{\nu}_h(t) \in \{0, 1\}$, where $\tilde{\nu}_h(t) = 1$ means working and $\tilde{\nu}_h(t) = 0$ means shirking. The principal therefore faces a moral hazard problem from the human agent. Conditional on no prior breakthrough, a human-generated breakthrough arrives according to a Poisson process with instantaneous rate $\lambda_h \tilde{\nu}_h(t)$, while an AI-generated breakthrough arrives according to an independent Poisson process with instantaneous rate $\lambda_a \nu_a(t)$. Therefore, conditional on survival up to time t , the total breakthrough arrival rate is $\lambda_h \tilde{\nu}_h(t) + \lambda_a \nu_a(t)$.

To activate the AI agent, the principal incurs a flow cost c_a whenever $\nu_a(t) = 1$. Likewise, to activate the human agent, the principal incurs a flow cost c_h whenever $\nu_h(t) = 1$. Whenever the principal does not activate the an agent (either the human agent or the AI agent), she can simply shut down the funding c_a or c_h so that the agents cannot work. The AI agent’s activity is publicly observable, while the human agent’s effort is unobservable, and shirking brings him a private flow benefit $\phi_h \in (0, c_h]$. Upon success, the principal observes which agent generated the breakthrough, collects a payoff R , makes a contractual transfer to the human agent, and the game ends. Both the principal and the human agent discount future payoffs at rate $r > 0$. Throughout the paper, similar to [Green and Taylor \(2016\)](#) and [Tian et al. \(2026\)](#), we assume that the human agent is beneficial to the principal, i.e., $\lambda_h R - c_h > \phi_h$; otherwise, the human agent would never be employed. For the AI agent, we discuss the impact of cost-ineffective AI agent, i.e., $\lambda_a R \leq c_a$, and cost-effective AI agent, i.e., $\lambda_a R > c_a$, in [Section 4](#).

At the beginning of the game, the principal, who aims to maximize her expected payoff, designs and commits to a contract $\Gamma := \{(\nu_i(t))_{t \geq 0}, (L(t))_{t \geq 0}, i \in \{a, h\}\}$, where $(\nu_h(t))_{t \geq 0}$ is the human

agent's *recommended* effort process, $(\nu_a(t))_{t \geq 0}$ is the AI agent's *designated* activity process, and $L(t)$ is the payment that the human agent receives if he generates the breakthrough at time t .

Agent's Problem Let $\nu_h = (\nu_h(s))_{s \geq 0}$ denote the human agent's recommended effort process, and define $\tilde{\nu}_h$ in a similar manner as the human agent's actual effort process. In addition, we let ν_a be the AI agent's designated activity process. We use τ_h and τ_a to denote the first arrival time of the human-generated and AI-generated breakthrough processes, respectively, and define the time of the first breakthrough as $\tau := \tau_h \wedge \tau_a$. We then define $W(t; (\tilde{\nu}_h, \nu); \Gamma)$ as the human agent's payoff-to-go at time t when his chosen effort is $\tilde{\nu}_h$, while the principal's recommended effort and designated processes are $\nu = (\nu_h, \nu_a)$ under contract Γ , given that the success has not arrived yet before time t . Then

$$W(t; (\tilde{\nu}_h, \nu); \Gamma) = \mathbb{E} \left[e^{-r(\tau_h - t)} L(\tau_h) \mathbb{1}\{\tau_h < \tau_a\} + \int_t^\tau e^{-r(s-t)} \phi_h \mathbb{1}\{\nu_h(s) = 1, \tilde{\nu}_h(s) = 0\} ds \mid \tau > t \right],$$

where the expectation is taken conditional on no breakthrough having occurred by time t , and with respect to the breakthrough processes induced by the human agent's actual effort process $\tilde{\nu}_h$ and the AI agent's designated activity process ν_a . In particular, at each time $s \geq t$, conditional on survival up to s , the total breakthrough arrival rate at time s is $\lambda_h \tilde{\nu}_h(s) + \lambda_a \nu_a(s)$. The first term inside the expectation is the human agent's discounted payment if he generates the breakthrough first, i.e., if $\tau_h < \tau_a$. The second term captures the agent's discounted private benefit from shirking: at any time s , he enjoys flow benefit ϕ_h whenever he is recommended to work, i.e., $\nu_h(s) = 1$, but instead chooses to shirk, $\tilde{\nu}_h(s) = 0$.

By the revelation principle, we can focus on direct mechanisms where the human agent has no incentive to deviate from the principal's recommended effort process ν_h , i.e.,

$$W(0; (\nu_h, \nu); \Gamma) \geq W(0; (\tilde{\nu}_h, \nu); \Gamma), \quad \forall \tilde{\nu}_h. \quad (\text{IC}^\circ)$$

For ease of notation, we use

$$w(t; \Gamma) := W(t; (\nu_h, \nu); \Gamma), \quad (1)$$

to denote the human agent's continuation payoff at time t when he follows the principal's recommendation under contract Γ . This continuation payoff is often called the "promised utility" in the dynamic contracting literature; see [Spear and Srivastava \(1987\)](#).

Assume that the human agent's outside option has value 0. Then the human agent is willing to participate in the game at each time t if the following individual rationality condition holds:

$$w(t; \Gamma) \geq 0, \quad \forall t \geq 0. \quad (\text{IR})$$

The following lemma provides an equivalent recursive representation of the [\(IC^o\)](#) constraint. In what follows, when the contract Γ is clear from the context, we simplify the notation to $w(t)$.

LEMMA 1. The function $w(t)$ as defined in (1) satisfies

$$\dot{w}(t) = (\lambda_a \nu_a(t) + \lambda_h \nu_h(t) + r) w(t) - \lambda_h \nu_h(t) L(t) \quad (2)$$

Furthermore, a contract Γ is incentive compatible, that is, (IC^o) holds if and only if

$$\nu_h(t) \left(L(t) - w(t) - \frac{\phi_h}{\lambda_h} \right) \geq 0. \quad (\text{IC}(t))$$

for any time $t \geq 0$.

Principal's Problem Under an incentive compatible contract Γ , the principal's total expected payoff, denoted by $P(\Gamma)$, is then:

$$P(\Gamma) := \mathbb{E} \left[e^{-r\tau} R - \int_0^\tau e^{-rs} (c_h \nu_h(s) + c_a \nu_a(s)) ds \right] - w(0; \Gamma),$$

which is the societal value generated by the project minus the human agent's total expected payoff under contract Γ . In particular, the first term, $\mathbb{E} [e^{-r\tau} R - \int_0^\tau e^{-rs} (c_h \nu_h(s) + c_a \nu_a(s)) ds]$ denotes the societal value generated by the project. Here, $e^{-r\tau} R$ is the discounted value of the breakthrough at time τ , regardless of whether it is generated by the human agent or the AI agent, while $\int_0^\tau e^{-rs} (c_h \nu_h(s) + c_a \nu_a(s)) ds$ captures the discounted operating costs incurred before the breakthrough.

It follows that the principal's optimization problem is:

$$\begin{aligned} P^* &:= \sup_{\Gamma} P(\Gamma) \\ \text{s.t. } & \text{(IC}^{\circ}) \text{ and (IR)}. \end{aligned} \quad (3)$$

Problem (3) is an infinite-dimensional optimization problem, as the principal chooses both a dynamic deployment policy and a dynamic compensation scheme over continuous time. Nevertheless, as we show in the next section, the optimal mechanism admits a simple characterization.

4. Main Results

In this section, we answer our main research question by characterizing when the principal should deploy the human agent alone, the AI agent alone, or both in a human–AI race, and how the human agent should be compensated upon a breakthrough. We begin with the case in which the AI agent is cost-ineffective, that is, $\lambda_a R \leq c_a$. In this region, the AI agent is available to the principal but does not contribute enough to justify its operating cost. The next proposition characterizes the optimal contract.

PROPOSITION 1 (Cost-Ineffective AI). *When the AI agent is cost-ineffective, i.e., $\lambda_a R \leq c_a$, the optimal contract works as follows: the principal never works with the AI agent. Instead, she only works with the human agent for a duration of*

$$T_H = \frac{1}{\lambda_h} \ln \left(\frac{\lambda_h R - c_h}{\phi_h} \right),$$

and the principal pays

$$L_H(t) = \frac{\phi_h}{\lambda_h} + \frac{\phi_h}{r} (1 - e^{-r(T_H - t)})$$

to the human agent if he generates success at $t \leq T_H$. If no success occurs before T_H , the contract terminates without payment.

Proposition 1 shows that when the AI agent does not have standalone positive value, i.e., $\lambda_a R \leq c_a$, the principal optimally leaves the AI agent idle and relies solely on the human agent. The optimal contract has a simple structure: the human agent is given a finite window to generate success, and the success payment declines over time. Since the AI agent is never used, the optimal contract coincides with the human-alone benchmark, in which the AI agent is unavailable at all. We denote the optimal contract in Proposition 1 by Γ_H^* .

Although the non-use of the AI agent is intuitive in this region, the reduction to the human-alone contract provides a useful benchmark for the more interesting cost-effective regime, $\lambda_a R > c_a$, on which we focus afterwards. At first glance, a cost-effective AI agent makes its persistent deployment compelling. The AI agent plays two roles. It accelerates breakthrough generation, creating positive standalone value $\lambda_a R - c_a$ and, through the threat of preemption, makes the human agent cheaper to incentivize. But as we argue in the next two subsections, this preemption may harm the principal in a human–AI race. In what follows, we first consider a benchmark in which the human agent faces persistent AI rivalry to build intuition, then relax the restriction of persistent AI rivalry and solve the principal’s optimal contract design problem.

4.1. Benchmark: Persistent AI Rivalry

We now consider a scenario in which the AI agent is deployed persistently. Once the contract starts, the AI agent remains active until a breakthrough occurs. The principal can decide whether and how long to keep the human agent in the race, and how to incentivize him. The following proposition presents the optimal contract.

PROPOSITION 2. *When the AI agent is cost-effective, i.e., $\lambda_a R - c_a \geq 0$, under the persistent AI rivalry scenario, the optimal contract works as follows: the principal works with the human agent for a duration of*

$$T_R = \left[\frac{1}{\lambda_h} \log \left(\frac{\lambda_h(rR + c_a) - c_h(r + \lambda_a)}{\phi_h(r + \lambda_a)} \right) \right]^+,$$

and the principal pays

$$L_R(t) = \frac{\phi_h}{\lambda_h} + \frac{\phi_h}{r + \lambda_a} \left(1 - e^{-(r + \lambda_a)(T_R - t)}\right),$$

to the human agent if he generates success at $t \leq T_R$. If no success occurs before T_R , the human agent is dropped, and the AI agent continues alone until success occurs.

Proposition 2 shows that when the AI agent is persistently active, it is optimal for the principal to start a race between the human agent and AI agent before a deadline T_R . If no success occurs before the deadline, then the principal resorts to the valuable AI agent alone afterwards. More specifically, if the AI agent becomes sufficiently effective, i.e., $\lambda_a \geq \frac{\lambda_h(rR + c_a)}{c_h + \phi} - r$, then $T_R = 0$. Here, the principal never employs the human agent, and the optimal contract collapses to AI-alone search from the outset, which is intuitive. If, instead, the AI agent is not sufficiently effective, i.e., $\lambda_a < \frac{\lambda_h(rR + c_a)}{c_h + \phi} - r$, then the human agent will be activated from the outset. The intuition is as follows: the human adds an additional breakthrough channel on top of the active AI agent, and this acceleration is more valuable earlier because the principal discounts future payoffs. Moreover, delaying the human agent's activation does not save incentive costs: conditional on no AI-generated breakthrough, the future contracting problem is the same except that payoffs are discounted. Hence, the principal either employs the human from the outset or does not employ him at all. We denote the optimal contract in Proposition 2 as Γ_R^* .

As discussed earlier, once the AI agent has positive standalone value, it is tempting to think that the principal should deploy the AI agent from the outset till a breakthrough occurs. An active AI agent adds an additional production channel and, by potentially preempting the human agent, can make the human agent easier to incentivize. This suggests that persistent AI agent rivalry should weakly improve upon the human-alone benchmark. The next proposition shows that this intuition is generally not true.

COROLLARY 1. *At the AI agent break-even boundary $\lambda_a R = c_a$, the optimal human-alone contract can strictly dominate the optimal contract under persistent AI agent rivalry for both the human agent and the principal, i.e.,*

$$w(0; \Gamma_H^*) > w(0; \Gamma_R^*), \quad P(\Gamma_H^*) > P(\Gamma_R^*).$$

Corollary 1 isolates the two effects of preemption by the AI agent in a boundary case. The first is an *incentive effect*. At the break-even boundary, $\lambda_a R = c_a$, the optimal deadline is the same under the human-alone contract and persistent AI agent rivalry, $T_H = T_R$, but the success payment is lower under persistent AI rivalry: $L_R(t) < L_H(t)$ for $t < T_R$. Because the human may be preempted by the AI agent, he is willing to work for a lower reward. AI rivalry thus makes the human agent cheaper to incentivize, and leaves him worse off.

The second is an *efficiency effect*. At the break-even boundary, the AI agent accelerates the resolution of the project but adds no standalone net surplus. Its preemption of the human agent therefore reduces the chance that society obtains the positive surplus generated by the human channel. From the principal's perspective, persistent AI rivalry trades off the incentive benefit from lower human rent against the efficiency loss from crowding out human-generated surplus. Corollary 1 shows that the efficiency loss can dominate the incentive benefit, making the principal strictly worse off.

This comparison conveys an important managerial implication: more competition is not necessarily better. AI agent rivalry can be double-edged. A persistent deployment of the AI agent can accelerate breakthrough generation and discipline the human agent, but when the AI agent contributes little net social surplus, persistent rivalry may reduce the payoffs of the human agent, the principal, and society as a whole. The principal should therefore balance the incentive benefit of AI agent rivalry against its efficiency loss. The next subsection studies the optimal way of using the AI agent as a rival to the human agent.

4.2. Optimal Dynamic Contract

We now allow the principal to dynamically activate the human agent and the AI agent over time. At each instant, the principal can rely on the human agent alone, the AI agent alone, both agents in a human–AI race, or neither agent. The next theorem characterizes the optimal contract.

THEOREM 1. *When $\lambda_a R > c_a$ and $r > 0$, let*

$$T_2^* = \left[\frac{1}{\lambda_h} \log \left(\frac{\lambda_h(rR + c_a) - c_h(r + \lambda_a)}{\phi_h(r + \lambda_a)} \right) \right]^+,$$

there exists $T_1^ \in [0, T_2^*]$ such that an optimal contract works as follows.*

- *Stage 1: Over time $t \in [0, T_1^*]$, only the human agent is active. If he generates the breakthrough at time $t \in [0, T_1^*]$, he receives payment*

$$L(t) = e^{-r(T_1^* - t)} \frac{\phi_h}{r + \lambda_a} \left(1 - e^{-(r + \lambda_a)(T_2^* - T_1^*)} \right) + \frac{\phi_h}{r} \left(1 - e^{-r(T_1^* - t)} \right) + \frac{\phi_h}{\lambda_h}.$$

- *Stage 2: Over time $t \in [T_1^*, T_2^*]$, both the human agent and the AI agent are active. If he generates the breakthrough at time $t \in [T_1^*, T_2^*]$, he receives payment*

$$L(t) = \frac{\phi_h}{r + \lambda_a} \left(1 - e^{-(r + \lambda_a)(T_2^* - t)} \right) + \frac{\phi_h}{\lambda_h}.$$

- *Stage 3: Over time $t \in [T_2^*, \infty)$, only the AI agent is active.*

If the breakthrough is generated by the AI agent, or if there is no breakthrough, the human agent receives no payment. The contract terminates immediately upon breakthrough.

In addition, given c_a , there exist thresholds $\check{\lambda}_a(c_a) < \hat{\lambda}_a(c_a)$ ¹ such that

¹ The expressions for $\check{\lambda}_a$, $\hat{\lambda}_a$ and the closed form of T_1^* and T_2^* are provided in Appendix A.5.

(i) If $\lambda_a \geq \hat{\lambda}_a(c_a)$, then

$$T_1^* = T_2^* = 0.$$

(ii) If $\check{\lambda}_a(c_a) \leq \lambda_a < \hat{\lambda}_a(c_a)$, then

$$T_2^* > T_1^* = 0.$$

(iii) If $\lambda_a < \check{\lambda}_a(c_a)$, then

$$T_2^* > T_1^* > 0.$$

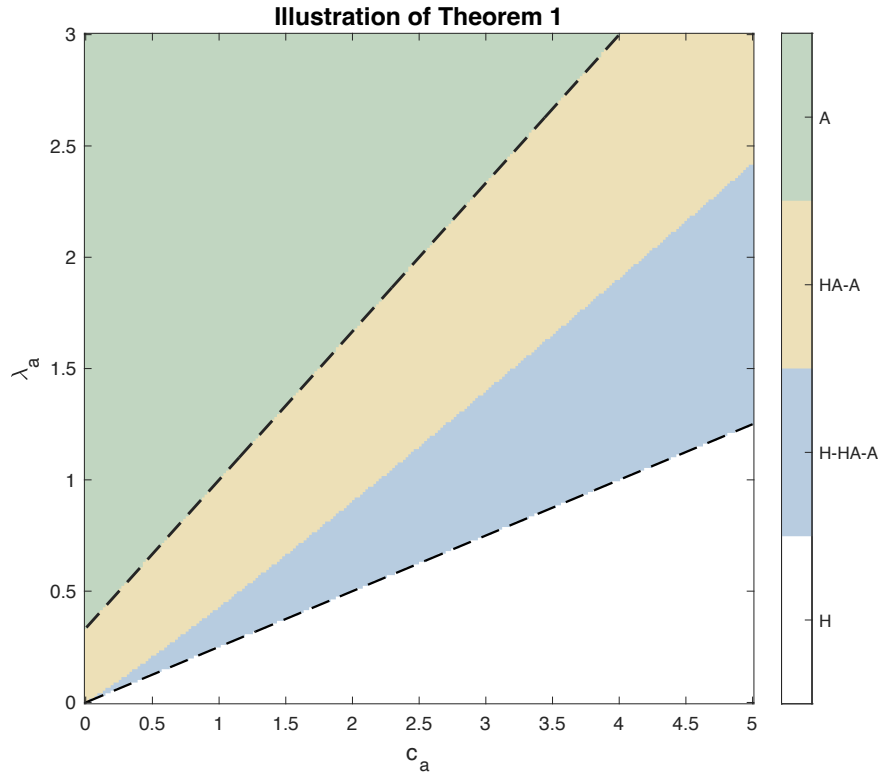
Theorem 1 shows that the optimal human–AI race does not necessarily begin immediately. Instead of keeping the AI agent active from the outset, the principal may first let the human work alone, then introduce the AI agent as a rival, and eventually rely on the AI agent alone. We denote the optimal contract in Theorem 1 as Γ_{HA}^* .

Consider first the case in which the AI agent is cost-effective but still relatively weak, $c_a/R < \lambda_a < \check{\lambda}_a(c_a)$ (blue region in Figure 1). Here, the optimal contract starts with a human-alone stage. This echoes Corollary 1. When the AI agent is relatively weak, the resulting efficiency loss can exceed the incentive-saving benefit from disciplining the human. Therefore, the principal initially frees the human agent from AI competition and lets the human work alone for a while.

Surprisingly, even near the break-even boundary, the principal does not simply retain the human agent until some cutoff and then switch to AI-alone search. Instead, after sufficient failure, she keeps the human agent active and introduces AI agent as a rival. The reason is that the two effects of AI preemption evolve over time. Early in the project, deploying the AI agent sacrifices too much of the potential high surplus generated by the human, i.e., a large efficiency loss. As failure persists, however, the remaining surplus of the human-alone path declines, so the efficiency loss from AI preemption becomes smaller. At the same time, the AI agent continues to provide incentive savings by disciplining the human agent, and, because $r > 0$, also creates acceleration value by bringing possible resolution forward. Once these benefits outweigh the shrinking efficiency loss, the principal starts the human–AI race. Since the AI agent is still relatively weak in this region, however, it is not optimal to rely on the AI agent alone immediately; the principal keeps the human in the race before eventually switching to AI-alone search.

As the AI agent becomes more effective, i.e., $\lambda_a \geq \check{\lambda}_a(c_a)$, the human-alone stage disappears ($T_1^* = 0$), and persistent deployment of the AI agent becomes globally optimal. This is because the effect created by AI preemption shifts in favor of early use: a more productive AI generates greater standalone value, imposes a smaller efficiency loss from preempting the human channel, and disciplines the human agent more strongly. Thus, the principal deploys the AI agent from the outset. For a moderately effective AI agent (the yellow region in Figure 1), this takes the form of an immediate human–AI race followed by AI-alone search; for a sufficiently effective AI agent (the

Figure 1 Theorem 1 Illustration



Notes. Model parameters: $R = 6$, $\lambda_h = 1.2$, $r = 0.2$, $c_h = 1 > \phi_h = 0.8$, $\lambda_a \in [0, 3]$, $c_a \in [0, 5]$. The labels indicate the decision regime over time: A means always AI-alone; HA-A means human-AI race at first, followed by AI-alone; H-HA-A means human-alone first, then human-AI race, and finally AI-alone; and H means always human-alone.

green region in Figure 1), the principal bypasses the human entirely and relies on the AI agent alone.

Theorem 1 therefore shows that the optimal use of AI agent competition is governed by the balance between the incentive and efficiency effects of preemption. When the AI agent is weakly effective, early preemption destroys too much human-channel value, so delayed competition is optimal. When the AI agent is more effective, the efficiency loss shrinks and the incentive and acceleration benefits grow, so the principal introduces the AI agent immediately. Thus, the optimal policy is not “more competition is always better.”

COROLLARY 2. *When the AI agent is not sufficiently efficient, $c_a/R < \lambda_a < \check{\lambda}_a$,*

$$w(0; \Gamma_{HA}^*) > w(0; \Gamma_R^*), \quad P(\Gamma_{HA}^*) > P(\Gamma_R^*).$$

The proof is straightforward and hence omitted. Corollary 2 strengthens Corollary 1. Corollary 2 shows that the issue is not the use of the AI agent as a rival, but the overly aggressive use of

AI rivalry. When the AI agent is weakly effective, a delayed human–AI race can make both the principal and the human agent strictly better off than keeping the AI agent persistently active.

Compared with the human-alone environment, a cost-effective AI agent expands the principal’s innovation capacity by adding a second discovery channel. In practice, however, the principal could also add another innovation channel by hiring another human agent. This raises a natural question: if the principal wants a second discovery channel, should it come from an AI agent or from another human agent? Moreover, how does this choice affect the focal human agent, who is the unique human agent in the human–AI organization and the representative human agent in the human–human organization?

5. Expanding Innovation Capacity: Human–AI versus Human–Human

In this section, we first formally introduce the human–human race where the second innovation channel is provided by another human agent. We then compare it with the optimal human–AI racing studied in Section 4.2 in which the second channel is provided by an AI agent. Our comparison focuses on the principal’s and a focal human agent’s payoff.

5.1. Human–Human Benchmark

In the human–human benchmark, a principal works with two symmetric human agents, indexed by $i = 1, 2$, to generate one innovation breakthrough. Time is continuous, and the project ends when the first breakthrough arrives. At each time $t \geq 0$, conditional on no prior breakthrough, the principal recommends whether each human agent should work. Let $\nu_i(t) \in \{0, 1\}, i = 1, 2$, denote the recommended effort process for agent i , where $\nu_i(t) = 1$ means that agent i is recommended to work at time t . The recommended effort is publicly observable, but actual effort is privately chosen. Let $\tilde{\nu}_i(t) \in \{0, 1\}, i = 1, 2$, denote agent i ’s actual effort, where $\tilde{\nu}_i(t) = 1$ means working and $\tilde{\nu}_i(t) = 0$ means shirking. Compared with the human–AI scenario, both agents in the human–human benchmark create moral hazard problems. If agent i works at time t , he generates a breakthrough according to an independent Poisson process with instantaneous rate λ_h . Thus, under actual effort profile $(\tilde{\nu}_1(t), \tilde{\nu}_2(t))$, the total breakthrough arrival rate is $\lambda_h(\tilde{\nu}_1(t) + \tilde{\nu}_2(t))$.

The principal pays a flow operating cost c_h for each active human agent. If agent i is recommended to work but shirks, he obtains private flow benefit $\phi_h \in (0, c_h]$. Upon a breakthrough, the principal observes which agent generated it, receives reward R , makes the contractual transfer, and the game ends. Both the principal and the agents discount future payoffs at rate $r > 0$.

At the beginning of the game, the principal designs and commits to a contract $\Gamma := \left\{ (\nu_i(t))_{t \geq 0}, (L_i(t))_{t \geq 0}, i = 1, 2 \right\}$, where $L_i(t)$ is the payment to agent i if agent i generates the first breakthrough at time t . The other agent receives no payment at that terminal time.

Let τ_i denote the first arrival time of agent i 's breakthrough process, and let $\tau := \tau_1 \wedge \tau_2$ be the project completion time. For $i \in \{1, 2\}$, let $\boldsymbol{\nu}_i = (\nu_i(s))_{s \geq 0}$ denote agent i 's recommended effort process, and define $\tilde{\boldsymbol{\nu}}_i$ analogously for actual effort. Given a contract Γ , agent i 's payoff-to-go at time t , conditional on no breakthrough before t , is

$$W_i^{HH}(t; (\tilde{\boldsymbol{\nu}}_i, \boldsymbol{\nu}); \Gamma) := \mathbb{E} \left[e^{-r(\tau_i - t)} L_i(\tau_i) \mathbf{1}\{\tau_i < \tau_j\} + \int_t^\tau e^{-r(s-t)} \phi_h \mathbf{1}\{\nu_i(s) = 1, \tilde{\nu}_i(s) = 0\} ds \mid \tau > t \right],$$

where $j \neq i$, and $\boldsymbol{\nu} = (\boldsymbol{\nu}_1, \boldsymbol{\nu}_2)$. The first term is agent i 's discounted payment if he generates the first breakthrough, and the second term is his discounted private benefit from shirking when he is recommended to work. Following [Shan \(2017\)](#), who studies a multi-agent moral hazard problem, we focus on Nash incentive-compatible contracts under which no agent wants to deviate unilaterally from the recommended effort process when the other agent follows his recommendation:

$$W_i^{HH}(0; (\boldsymbol{\nu}_i, \boldsymbol{\nu}); \Gamma) \geq W_i^{HH}(0; (\tilde{\boldsymbol{\nu}}_i, \boldsymbol{\nu}); \Gamma), \quad \forall \tilde{\boldsymbol{\nu}}_i, \quad i = 1, 2. \quad (\text{IC}_i^{HH})$$

Let

$$w_i^{HH}(t; \Gamma) := W_i^{HH}(t; (\boldsymbol{\nu}_i, \boldsymbol{\nu}); \Gamma)$$

denote agent i 's continuation payoff when both agents follow the contract. The individual rationality constraint is

$$w_i^{HH}(t; \Gamma) \geq 0, \quad t \geq 0, \quad i = 1, 2. \quad (\text{IR}_i)$$

The principal's expected payoff under a contract Γ is the total discounted surplus generated by the project net of the agents' equilibrium rents:

$$P^{HH}(\Gamma) := \mathbb{E} \left[e^{-r\tau} R - \int_0^\tau e^{-rs} c_h(\nu_1(s) + \nu_2(s)) ds \right] - w_1^{HH}(0; \Gamma) - w_2^{HH}(0; \Gamma).$$

The human-human contracting problem is

$$\sup_{\Gamma} P^{HH}(\Gamma) \quad \text{subject to } (\text{IC}_i^{HH}), \text{ and } (\text{IR}_i), i = 1, 2.$$

Before solving this problem, we note that the human-human benchmark is closely related to the two-agent case studied in [Tian et al. \(2026\)](#), but differs in two important respects. First, in our model, the agents' efforts are additive: when both agents work, their breakthrough rates add up. In contrast, [Tian et al. \(2026\)](#) study an environment in which agents' efforts are substitutable. Second, we allow for discounting, whereas they assume no discounting. These differences are not cosmetic and lead to fundamentally different optimal contracts. [Tian et al. \(2026\)](#) show that it is asymptotically optimal for the principal to rotate between the two agents frequently. In our setting, by contrast, the following proposition shows that the optimal contract has the two human agents work simultaneously.

PROPOSITION 3. *In the human–human setting, the optimal contract recommends that both human agents work until time*

$$T_{HH} = \frac{1}{\lambda_h} \ln \left(\frac{\lambda_h R - c_h}{\phi_h} \right),$$

and the principal pays

$$L_i(t) = \frac{\phi_h}{\lambda_h} + \frac{\phi_h}{\lambda_h + r} (1 - e^{-(\lambda_h + r)(T_{HH} - t)}), \quad i = 1, 2, \quad 0 \leq t \leq T_{HH},$$

to the successful agent, while the other agent receives no payment. If no breakthrough arrives by time T_{HH} , the contract terminates without payment.

Proposition 3 shows that in the human–human benchmark, the optimal contract induces both human agents to work simultaneously until a common endogenous deadline, and rewards only the agent who generates the first breakthrough. We use Γ_{HH}^* to denote the optimal contract in Proposition 3. For ease of reference in the next subsection, we let $P_{HH}^* = P^{HH}(\Gamma_{HH}^*)$, and $w_{HH}^* = w_1^{HH}(0; \Gamma_{HH}^*)$ where symmetry implies $w_1^{HH}(0; \Gamma_{HH}^*) = w_2^{HH}(0; \Gamma_{HH}^*)$.

5.2. Comparison: Human–AI versus Human–Human

Before comparing the human–AI and human–human organizations directly, it is useful to study how the human–AI organization changes with the AI agent’s capability λ_a . We focus on λ_a because it directly captures the effectiveness of the AI agent, while the human–human benchmark is independent of λ_a . This makes the comparison between the two environments clean.

PROPOSITION 4. *Under the human–AI environment, the principal’s payoff under the optimal contract, i.e., $P(\Gamma_{HA}^*(\lambda_a))$, is increasing in λ_a , while the agent’s payoff under the optimal contract, i.e., $w(0; \Gamma_{HA}^*(\lambda_a))$, is decreasing in λ_a .*

Proposition 4 shows that the AI agent’s capability has opposite effects on the two parties. A more capable AI agent is more attractive to the principal as it raises the arrival rate of breakthroughs and strengthens the human agent’s incentives through a greater threat of preemption. The second force exactly hurts the human agent, because stronger AI agent competition reduces his chance of generating the first breakthrough and lowers the rent needed to induce effort.

Motivated by Proposition 4, define two indifference thresholds:

$$\lambda_a^P := \sup\{\lambda_a : P(\Gamma_{HA}^*(\lambda_a)) \leq P_{HH}^*\}, \text{ and } \lambda_a^A := \sup\{\lambda_a : w(0; \Gamma_{HA}^*(\lambda_a)) \geq w_{HH}^*\} \quad (4)$$

The threshold λ_a^P is the AI agent’s capability level at which the principal switches from preferring the human–human organization to preferring the human–AI organization. The threshold λ_a^A is the AI capability level at which the focal human agent switches from preferring the human–AI

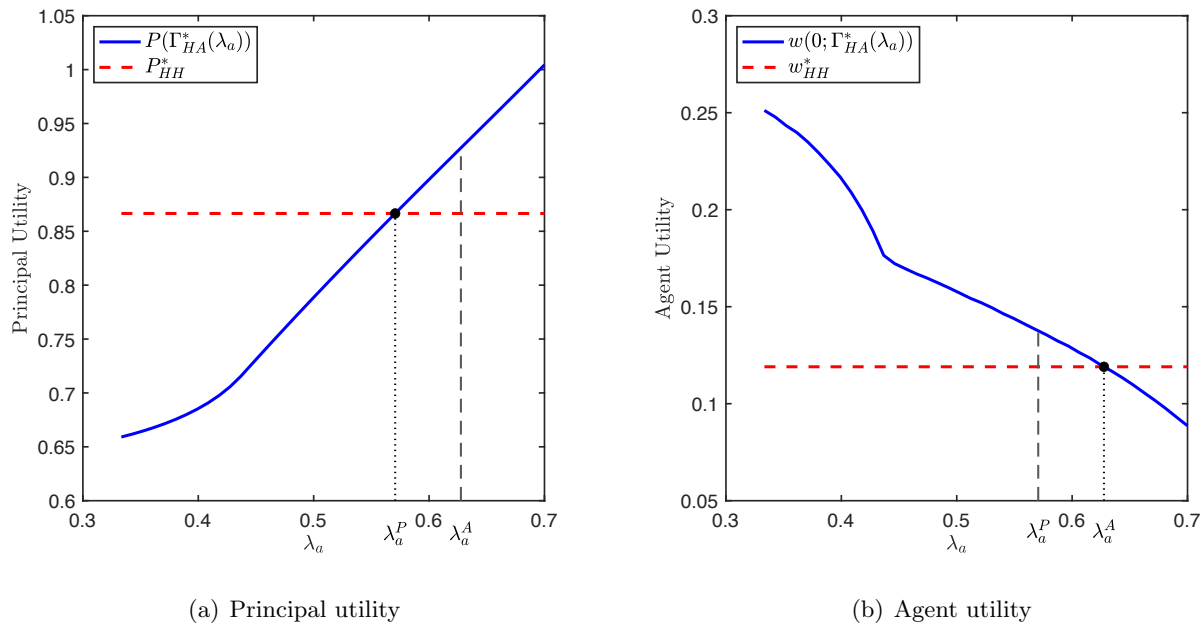
organization to preferring the human–human organization. The ranking of these two thresholds determines whether a moderately capable AI agent creates joint preference for human–AI adoption or joint opposition to it. The next result shows that this ranking can be governed by the AI agent’s operating cost.

PROPOSITION 5. *There exists a threshold c_a^* such that*

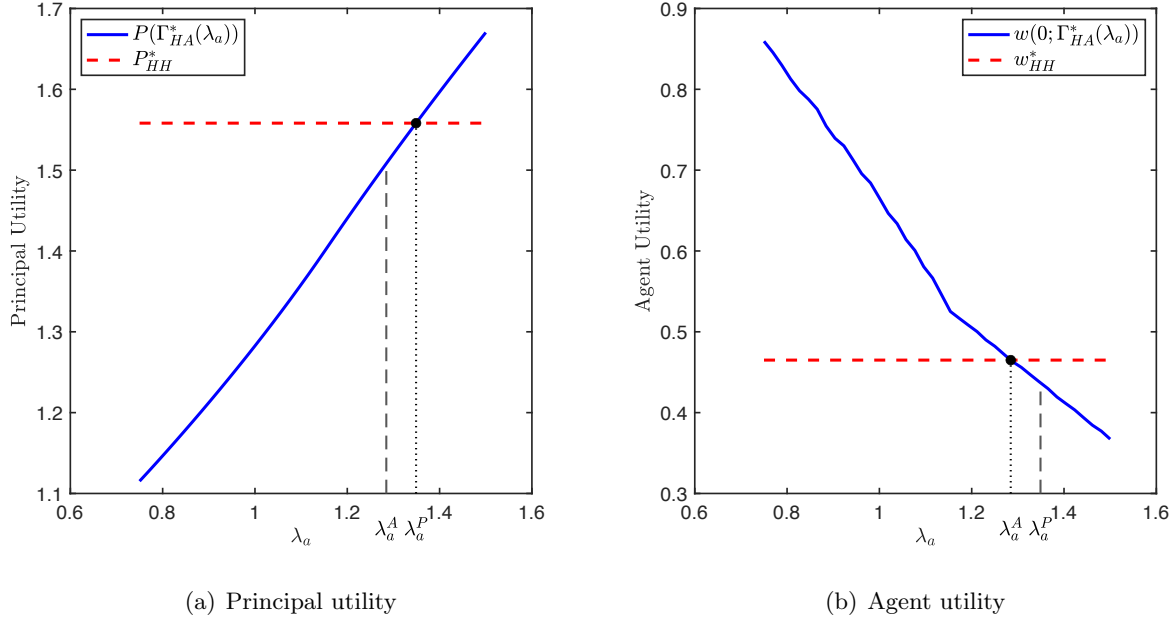
$$c_a \leq c_a^* \iff \lambda_a^P \leq \lambda_a^A.$$

When the AI agent’s operating cost is low (see Figure 2), the principal can benefit from the AI agent even when the AI agent is only moderately capable. At such capability levels, the AI agent is not yet strong enough to impose more competitive pressure on the focal human agent than another human would. Hence $\lambda_a^P \leq \lambda_a^A$. By contrast, when the AI agent’s operating cost is high (see Figure 3), the AI agent must be highly capable so that the human–AI organization becomes attractive to the principal. But by the time the AI agent is capable enough to compensate for its high cost, it also becomes a strong competitor to the focal human agent. Hence $\lambda_a^P > \lambda_a^A$. The following corollary shows how the AI agent’s ability shapes the preferences of the principal and the focal human agent between the human–human and human–AI organizations.

Figure 2 Small AI agent’s flow cost: comparison of principal and focal human agent payoffs



Notes. Model parameters: $\lambda_h = 4/3$, $r = 4/9$, $R = 3$, $c_a = 1$, $c_h = 2.2667 > \phi_h = 0.22667$, $\lambda_a \in [c_a/R, 0.7]$.

Figure 3 Large AI agent's flow cost: comparison of principal and focal human agent payoffs

Notes. Model parameters: $\lambda_h = 10/3$, $r = 0.5$, $R = 4$, $c_a = 3$, $c_h = 4 > \phi_h = 2.2$, $\lambda_a \in [c_a/R, 1.5]$.

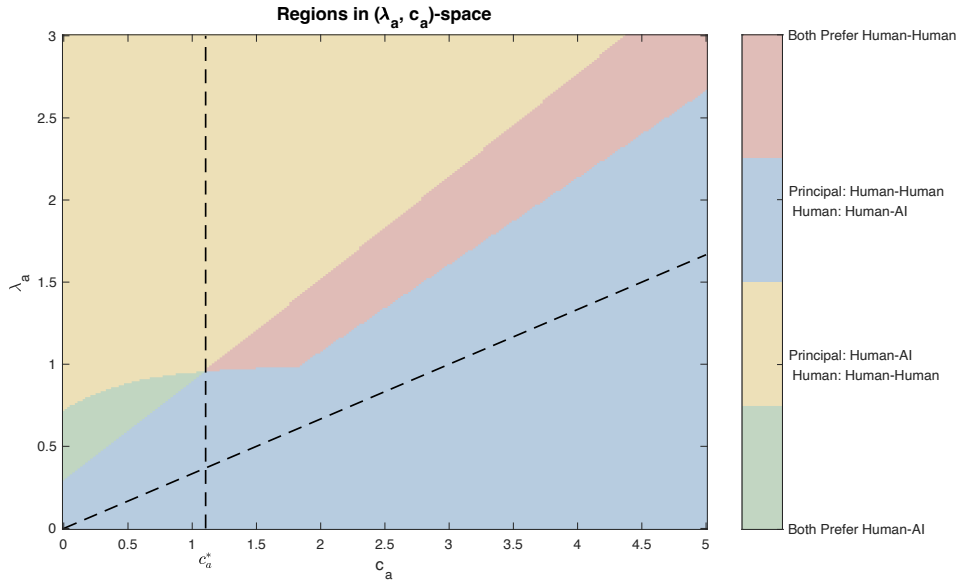
COROLLARY 3. Let

$$\underline{\lambda}_a := \min\{\lambda_a^P, \lambda_a^A\}, \quad \bar{\lambda}_a := \max\{\lambda_a^P, \lambda_a^A\}.$$

Then the preference comparison between the human-AI and human-human organizations is as follows.

- (a) **Weak AI Agent** ($\lambda_a < \underline{\lambda}_a$): the principal prefers the human-human organization, while the focal human agent prefers the human-AI organization.
- (b) **Strong AI Agent** ($\lambda_a > \bar{\lambda}_a$): the principal prefers the human-AI organization, while the focal human agent prefers the human-human organization. Thus, their preferences conflict.
- (c) **Intermediate AI Agent** ($\underline{\lambda}_a < \lambda_a < \bar{\lambda}_a$): the two parties agree. If $c_a \leq c_a^*$, then both prefer the human-AI organization. If $c_a > c_a^*$, then both prefer the human-human organization.

Corollary 3 is illustrated in Figure 4. When the AI agent is extremely weak or strong, the preferences of the principal and the focal human agent conflict. In particular, when the AI agent is weakly cost-effective (see the blue region in Figure 4), it is not attractive enough for the principal but is relatively benign for the focal human agent. When the AI agent is strongly cost-effective (see the yellow region in Figure 4), it becomes attractive to the principal but too threatening for the focal human agent. Preferences align in the intermediate region. A low-cost AI agent (see the green region in Figure 4) makes both parties prefer the human-AI organization to the human-human organization, whereas a high-cost AI agent (see the red region in Figure 4) creates a joint preference for adding a second human agent.

Figure 4 Principal and agent's preference between human–AI and human–human organization

Notes. Model parameters: $R = 3, \lambda_h = 1, r = 0.6, c_h = 1 > \phi_h = 0.1, c_a \in [0, 5], \lambda_a \in [0, 3]$.

6. Imperfect AI Agent with Inspection

In the base model, we assume that outcomes generated by the AI agent are genuine breakthroughs and therefore yield reward R to the principal. In practice, however, AI-generated outcomes may be spurious and fail to produce value. In this section, we examine an extension in which each AI-generated outcome is a true innovation with probability $p > 0$ and a fake breakthrough with probability $1 - p$.²

Upon observing an AI-generated outcome at time t , the principal chooses an inspection decision $\kappa(t) \in \{0, 1\}$, where $\kappa(t) = 1$ means that the principal inspects the outcome and $\kappa(t) = 0$ means that she does not. Inspection is perfect: after inspection, the principal observes whether the AI-generated outcome is true or fake. If the principal does not inspect, she accepts the AI-generated outcome, receives reward R if it is true and 0 if it is fake, and the process ends. If the principal inspects, she pays a fixed cost K . When the outcome is true, the principal receives reward R and the process ends; when the outcome is fake, the principal receives no reward and the process continues.

In this extension, we impose two assumptions. First, the AI agent is cost-effective in the sense that its expected benefit exceeds its flow cost, i.e., $p\lambda_a R - c_a > 0$. Second, we focus on the case in which inspection is sufficiently costly, i.e., $K \geq K^* := (1 - p)(\lambda_a p R - c_a) / (r + \lambda_a)$. This assumption captures settings in which verification is costly relative to the value of simply accepting AI-generated

² When $p = 1$, the extension reduces to the base model in Section 3.

outcomes. It makes the inspection decision nontrivial; otherwise, the principal would always inspect AI-generated outcomes.

For consistency, we reuse the notation from Section 3. Similar to the base model, the principal designs and commits to a contract Γ . In addition to the human agent's recommended effort process $(\nu_h(t))_{t \geq 0}$, the AI agent's designated activity process $(\nu_a(t))_{t \geq 0}$, and the payment $L(t)$ to the human agent upon a human-generated breakthrough, the principal also commits to an inspection rule $(\kappa(t))_{t \geq 0}$ for AI-generated outcomes. Mathematically,

$$\Gamma := \{(\nu_i(t))_{t \geq 0}, (L(t))_{t \geq 0}, (\kappa(t))_{t \geq 0} : i \in \{a, h\}\}.$$

We use τ_h to denote the first arrival time of the human-generated breakthrough, and τ_a to denote the AI-induced termination time, i.e., the first AI-generated outcome after which the process ends. This can occur either when an AI-generated outcome arrives and the principal chooses not to inspect it, or when an AI-generated outcome arrives, the principal inspects it, and the outcome is a true innovation. If the inspected AI-generated outcome is fake, the process continues. We then define the terminal time of the process as $\tau := \tau_h \wedge \tau_a$.

The human agent's payoff-to-go at time t , when he chooses effort process $\tilde{\nu}_h$ while the principal recommends effort process $\nu = (\nu_a, \nu_h)$ under contract Γ , conditional on the process not having ended by time t , is

$$W(t; (\tilde{\nu}_h, \nu); \Gamma) = \mathbb{E} \left[e^{-r(\tau_h - t)} L(\tau_h) \mathbb{1}\{\tau_h < \tau_a\} + \int_t^\tau e^{-r(s-t)} \phi_h \mathbb{1}\{\nu_h(s) = 1, \tilde{\nu}_h(s) = 0\} ds \mid \tau > t \right].$$

Similar to the main model, we focus on direct mechanisms in which the human agent has no incentive to deviate from the principal's recommended effort process ν_h , i.e.,

$$W(0; (\nu_h, \nu); \Gamma) \geq W(0; (\tilde{\nu}_h, \nu); \Gamma), \quad \forall \tilde{\nu}_h. \quad (\text{IC})$$

We use

$$w(t; \Gamma) := W(t; (\nu_h, \nu); \Gamma), \quad (5)$$

to denote the human agent's continuation payoff at time t when he follows the principal's recommendation under a contract Γ .

The agent is willing to participate at each time t if the following individual rationality condition holds:

$$w(t; \Gamma) \geq 0, \quad \forall t \geq 0. \quad (\text{IR})$$

To evaluate the principal's inspection cost, we denote $N_a(t)$ to be the counting process of AI-generated outcomes by time t . Then under an incentive-compatible contract Γ , the principal's total expected payoff, denoted by $P(\Gamma)$, is

$$P(\Gamma) := \mathbb{E} \left[e^{-r\tau} (R \mathbb{1}\{\tau_h < \tau_a\} + pR \mathbb{1}\{\tau_a < \tau_h, \kappa(\tau_a) = 0\} + R \mathbb{1}\{\tau_a < \tau_h, \kappa(\tau_a) = 1\}) - \int_0^\tau e^{-rt} (c_a \nu_a(t) + c_h \nu_h(t)) dt - K \int_0^\tau e^{-rt} \kappa(t) dN_a(t) \right] - w(0; \Gamma). \quad (6)$$

The principal's optimization problem is then

$$\sup_{\Gamma} P(\Gamma), \quad \text{subject to } \text{\textcolor{red}{(IC)}} \text{ and } \text{\textcolor{red}{(IR)}}.$$

Relative to the base model, the principal now faces both a dynamic deployment problem and a dynamic verification problem. She chooses whether to rely on the human agent alone, the AI agent alone, or both agents in a human–AI race, how to compensate the human agent, and whether to inspect AI-generated outcomes.

Next, we define a class of contracts that includes both dynamic deployment and dynamic verification.

DEFINITION 1 (CONTRACT $\Gamma_{H-AC-A}(T_1, T_2, T_3)$). Given switching dates $0 \leq T_1 \leq T_2 \leq T_3 \leq \infty$, the contract works as follows:

- Stage 1: Over time $t \in [0, T_1]$, only the human agent is active. If he generates the breakthrough at time $t \in [0, T_1]$, he receives payment

$$L(t) = e^{-r(T_1-t)} \left[e^{-(r+p\lambda_a)(T_2-T_1)} \frac{\phi_h}{r+\lambda_a} \left(1 - e^{-(r+\lambda_a)(T_3-T_2)} \right) + \frac{\phi_h}{r+p\lambda_a} \left(1 - e^{-(r+p\lambda_a)(T_2-T_1)} \right) \right] + \frac{\phi_h}{r} \left(1 - e^{-r(T_1-t)} \right) + \frac{\phi_h}{\lambda_h}.$$

- Stage 2: Over time $t \in [T_1, T_2]$, both the human agent and the AI agent are active, and the principal inspects AI-generated outcomes. If the human agent generates the breakthrough at time $t \in [T_1, T_2]$, he receives payment

$$L(t) = e^{-(r+p\lambda_a)(T_2-t)} \frac{\phi_h}{r+\lambda_a} \left(1 - e^{-(r+\lambda_a)(T_3-T_2)} \right) + \frac{\phi_h}{r+p\lambda_a} \left(1 - e^{-(r+p\lambda_a)(T_2-t)} \right) + \frac{\phi_h}{\lambda_h}.$$

- Stage 3: Over time $t \in [T_2, T_3]$, both the human agent and the AI agent are active, and the principal does not inspect AI-generated outcomes. If the human agent generates the breakthrough at time $t \in [T_2, T_3]$, he receives payment

$$L(t) = \frac{\phi_h}{r+\lambda_a} \left(1 - e^{-(r+\lambda_a)(T_3-t)} \right) + \frac{\phi_h}{\lambda_h}.$$

- Stage 4: Over time $t \in [T_3, \infty)$, only the AI agent is active, and the principal does not inspect AI-generated outcomes.

If the terminal outcome is induced by AI, or if no breakthrough occurs, the human agent receives no payment. $\square$

The contract class Γ_{H-AC-A} nests the optimal contract in Theorem 1 as a special case. Relative to the base model, it allows the human–AI racing stage to be split into two substages: an inspected racing stage, in which AI-generated outcomes are verified, and an uninspected racing stage, in which they are accepted without verification. The next theorem shows that an optimal contract belongs to this class.

THEOREM 2. *With imperfect AI, given $K \geq K^*$, there always exists $0 \leq T_1^* \leq T_2^* \leq T_3^*$ such that an optimal contract is $\Gamma_{H-AC-A}(T_1^*, T_2^*, T_3^*)$.*

The logic of Theorem 2 mirrors the optimal contract in the base model, but the imperfect AI agent adds an additional reason to delay its deployment. In the base model, early use of the AI agent can be costly because a genuine AI-generated breakthrough may preempt a high-surplus human path. With imperfect AI agent, the principal also faces the risk of false termination: an AI-generated outcome may be fake, and accepting it ends the project without a genuine breakthrough. Thus, early in the project, the principal is even more reluctant to introduce AI agent competition.

If failure persists, however, human–AI racing becomes attractive. As in the base model, the AI agent serves two roles: it raises the arrival rate of candidate breakthroughs and strengthens the human agent’s incentives by competing for the breakthrough. With the imperfect AI agent, the principal must also decide whether to verify AI-generated outcomes. At the beginning of racing, inspection is valuable because the continuation value of the project remains high. Rejecting a fake AI-generated outcome preserves the possibility of obtaining a genuine breakthrough later, either from the human agent or from the AI agent.

Over time, discounting reduces the value of keeping the project alive. Eventually, the benefit of avoiding premature termination no longer justifies the inspection cost. The principal then continues to use the AI agent as both a productive channel and an incentive device, but stops inspecting its outputs.

Thus, imperfect AI agent does not overturn the delayed-racing logic of the base model. Rather, it refines the human–AI racing stages: with the imperfect AI agent, the racing stage starts with inspected racing and then transitions to uninspected racing.

7. Conclusion

We study a dynamic contracting problem in which a principal seeks an innovation breakthrough by choosing which productive channels to activate: a strategic human agent, a non-strategic AI

agent, or both in a human–AI race. The human agent may be highly productive, but his effort is privately chosen and must be incentivized. The AI agent, by contrast, does not create a moral hazard problem, but it is costly to operate and can independently generate a breakthrough that preempts the human agent.

Our analysis shows that the AI agent affects optimal contracting through two distinct channels. First, the AI agent expands the principal’s innovation capacity by creating an additional source of breakthroughs. Second, the AI agent changes the human agent’s incentives by competing for the breakthrough and thereby reducing the rent needed to induce effort. This incentive benefit, however, may come with an efficiency cost: early deployment of the AI agent can prematurely preempt a valuable human innovation path. As a result, a cost-effective AI agent need not be deployed from the outset. The optimal mechanism may instead begin with human-alone search, then introduce the AI agent as a rival, and eventually switch to AI-alone search. As the AI agent’s capability rises, the initial human-alone phase vanishes and, beyond a further threshold, the principal relies on the AI agent alone throughout.

We further study how the AI agent compares with another common way of expanding innovation capacity: hiring a second human agent. Relative to a human–human benchmark, the principal favors a more capable AI agent, while the focal human agent fears the stronger competition it creates. As a result, the two parties’ preferences align only for intermediate AI agent capability, and the direction of that agreement is governed by the AI agent’s operating cost. A low-cost, moderately capable AI agent can make both the firm and the focal human agent better off relative to hiring another human agent, whereas a costly, moderately capable AI agent can make both worse off.

Finally, we extend the model to the imperfect AI agent whose outputs may be spurious and costly to verify. The imperfect AI agent strengthens the motive to delay the deployment of the AI agent, because early use of the AI agent may terminate the project falsely and preempt a valuable human path. Once the AI agent is introduced, the optimal mechanism may initially inspect AI-generated outcomes, and later stop inspecting. Thus, imperfect AI agent does not overturn the basic dynamic structure of delayed racing; instead, it refines the racing stage through time-varying verification.

Taken together, our results offer guidance for firms organizing human–AI innovation. Aggressive AI agent competition is not always better. Firms should treat competition from the AI agent as a disciplining instrument whose use should be matched to its productive value, weigh the AI agent against additional human talent based on both capability and cost, and concentrate verification of AI-generated outcomes in the early, high-stakes stages of a project.

Several directions remain open for future research. One is to allow for richer interactions between human and AI agent channels, including complementarity, learning, or switching costs. Another is to examine settings with multiple human agents or multiple AI systems competing or collaborating within the same project.

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Appendix A: Analytical Proofs

Throughout the appendix we write

$$\psi_a(t) := \int_0^t \nu_a(s) ds, \quad \psi_h(t) := \int_0^t \nu_h(s) ds,$$

and

$$S_A := \frac{\lambda_a R - c_a}{r + \lambda_a}, \quad S_{AH} := \frac{(\lambda_a + \lambda_h)R - (c_a + c_h)}{r + \lambda_a + \lambda_h}, \quad S_H := \frac{\lambda_h R - c_h}{r + \lambda_h}, \quad (\text{A.1})$$

the first-best social surpluses from operating the AI alone, both agents, and the human alone throughout, respectively.

A.1. Proof of Lemma 1

Under our model the human agent can shirk when recommended to work but cannot work when inactive, i.e., any actual effort process satisfies $\tilde{\nu}_h(t) \leq \nu_h(t)$ for all $t \geq 0$. Given $(\tilde{\nu}_h, \nu_a)$, the human agent's continuation payoff from time t onward is

$$W(t; (\tilde{\nu}_h, \nu_a); \Gamma) = \int_t^\infty e^{-\int_t^s (r + \lambda_a \nu_a(u) + \lambda_h \tilde{\nu}_h(u)) du} \left[\lambda_h \tilde{\nu}_h(s) L(s) + \phi_h(\nu_h(s) - \tilde{\nu}_h(s)) \right] ds, \quad (\text{A.2})$$

and under obedience,

$$w(t) = W(t; (\nu_h, \nu_a); \Gamma) = \int_t^\infty e^{-\int_t^s (r + \lambda_a \nu_a(u) + \lambda_h \nu_h(u)) du} \lambda_h \nu_h(s) L(s) ds. \quad (\text{A.3})$$

Differentiating (A.3) by Leibniz' rule yields

$$\dot{w}(t) = (r + \lambda_a \nu_a(t) + \lambda_h \nu_h(t)) w(t) - \lambda_h \nu_h(t) L(t),$$

which proves (2).

For the second part, note first that for any deviation $\tilde{\nu}_h$ that follows the recommendation before t , i.e.,

$$\tilde{\nu}_h^t(s) = \nu_h(s), \text{ if } s < t; \text{ and } \tilde{\nu}_h^t(s) = \hat{\nu}_h(s), \text{ if } s \geq t.$$

We have

$$W(0; (\nu_h, \nu_a); \Gamma) - W(0; (\tilde{\nu}_h, \nu_a); \Gamma) = e^{-\int_0^t (r + \lambda_a \nu_a(u) + \lambda_h \nu_h(u)) du} \left[w(t) - W(t; (\tilde{\nu}_h, \nu_a); \Gamma) \right], \quad (\text{A.4})$$

so (IC^o) implies the continuation-IC property

$$w(t) \geq W(t; (\tilde{\nu}_h, \nu_a); \Gamma), \quad \forall t, \forall \tilde{\nu}_h. \quad (\text{A.5})$$

We now prove necessity of (IC(t)). Fix $t \geq 0$, $\delta > 0$, and consider the deviation $\tilde{\nu}_h^{t, \delta}$ that shirks on $[t, t + \delta]$ and obeys otherwise, that is,

$$\tilde{\nu}_h^{t, \delta}(s) = \begin{cases} 0, & s \in [t, t + \delta], \\ \nu_h(s), & o.w. \end{cases}$$

Write $B_t(s) := \int_t^s (r + \lambda_a \nu_a(u)) du$ and $W^{t, \delta}(t) := W(t; (\tilde{\nu}_h^{t, \delta}, \nu_a); \Gamma)$; then

$$W^{t, \delta}(t) = \int_t^{t+\delta} e^{-B_t(s)} \phi_h \nu_h(s) ds + e^{-B_t(t+\delta)} w(t + \delta). \quad (\text{A.6})$$

By (2), $\frac{d}{ds} (e^{-B_t(s)} w(s)) = -e^{-B_t(s)} \lambda_h \nu_h(s) (L(s) - w(s))$, so

$$w(t) = e^{-B_t(t+\delta)} w(t + \delta) + \int_t^{t+\delta} e^{-B_t(s)} \lambda_h \nu_h(s) (L(s) - w(s)) ds. \quad (\text{A.7})$$

Subtracting (A.6) from (A.7) and using (A.5),

$$0 \leq w(t) - W^{t, \delta}(t) = \int_t^{t+\delta} e^{-B_t(s)} \nu_h(s) [\lambda_h (L(s) - w(s)) - \phi_h] ds.$$

It follows that

$$\lim_{\delta \rightarrow 0} \frac{w(t) - W^{t,\delta}(t)}{\delta} = \nu_h(t) [\lambda_h(L(t) - w(t)) - \phi_h] \geq 0.$$

where the equality comes from L'Hôpital's Rule. This proves necessity.

It remains to prove sufficiency. Suppose (IC(t)) holds for all t , and fix any $\tilde{\nu}_h$. With $Q_t(s) := \int_t^s (r + \lambda_a \nu_a + \lambda_h \tilde{\nu}_h) du$, (2) gives

$$\frac{d}{ds} \left(e^{-Q_t(s)} w(s) \right) = -e^{-Q_t(s)} \left[\lambda_h \tilde{\nu}_h(s) L(s) + \lambda_h (\nu_h(s) - \tilde{\nu}_h(s)) (L(s) - w(s)) \right].$$

Integrating from t to T , we obtain

$$w(t) = e^{-Q_t(T)} w(T) + \int_t^T e^{-Q_t(s)} \left[\lambda_h \tilde{\nu}_h(s) L(s) + \lambda_h (\nu_h(s) - \tilde{\nu}_h(s)) (L(s) - w(s)) \right] ds. \quad (\text{A.8})$$

Because $Q_t(T) \geq r(T-t)$ and w is bounded whenever L is bounded, we have

$$\lim_{T \rightarrow \infty} e^{-Q_t(T)} w(T) = 0.$$

Letting $T \rightarrow \infty$ in (A.8), we have

$$w(t) = \int_t^\infty e^{-Q_t(s)} \left[\lambda_h \tilde{\nu}_h(s) L(s) + \lambda_h (\nu_h(s) - \tilde{\nu}_h(s)) (L(s) - w(s)) \right] ds. \quad (\text{A.9})$$

Subtracting (A.2) from (A.9),

$$w(t) - W(t; (\tilde{\nu}_h, \nu_a); \Gamma) = \int_t^\infty e^{-Q_t(s)} (\nu_h(s) - \tilde{\nu}_h(s)) \left[\lambda_h (L(s) - w(s)) - \phi_h \right] ds. \quad (\text{A.10})$$

Whenever $\nu_h(s) - \tilde{\nu}_h(s) > 0$ we have $\nu_h(s) = 1$, so (IC(t)) makes the integrand nonnegative. Hence $w(t) \geq W(t; (\tilde{\nu}_h, \nu_a); \Gamma)$ for all t and all $\tilde{\nu}_h$; taking $t = 0$ shows Γ is incentive compatible. Q.E.D.

A.2. Proof of Proposition 1

We first derive a revenue upper bound and then show that the proposed contract is feasible and attains it.

LEMMA A1 (Human agent promised utility lower bound). *For any incentive-compatible and individually rational contract Γ ,*

$$w(0; \Gamma) \geq \underline{w}(0; \Gamma) := \phi_h \int_0^\infty \nu_h(t) e^{-rt - \lambda_a \psi_a(t)} dt.$$

Proof: See Appendix D.1 Q.E.D.

Step 1: Revenue upper bound. For any contract Γ , the societal value is

$$V(\Gamma) = \int_0^\infty e^{-rt - \lambda_h \psi_h(t) - \lambda_a \psi_a(t)} ((\lambda_h R - c_h) \nu_h(t) + (\lambda_a R - c_a) \nu_a(t)) dt,$$

so by Lemma A1 the principal's payoff

$$P(\Gamma) = V(\Gamma) - w(0; \Gamma) \leq U(\nu_h, \nu_a) := \int_0^\infty e^{-rt - \lambda_a \psi_a(t)} \left(((\lambda_h R - c_h) e^{-\lambda_h \psi_h(t)} - \phi_h) \nu_h(t) + (\lambda_a R - c_a) e^{-\lambda_h \psi_h(t)} \nu_a(t) \right) dt.$$

We solve

$$\bar{P} := \max_{\nu_h(t) \in \{0,1\}, \nu_a \in \{0,1\}, \forall t} U(\nu_h, \nu_a). \quad (\text{A.11})$$

Let

$$\bar{T}_h := \frac{1}{\lambda_h} \ln \frac{\lambda_h R - c_h}{\phi_h} > 0.$$

We first show that under an optimal solution ν_a^*, ν_h^* to problem (A.11), $\psi_h^*(\infty) = \int_0^\infty \nu_h^*(t) dt \leq \bar{T}_h$ by contradiction. Suppose $\psi_h^*(\infty) > \bar{T}_h$. Let s be the first time at which $\psi_h^*(t)$ reaches $\bar{T}_h$, i.e., $s = \inf\{t : \psi_h^*(t) = \bar{T}_h\}$, then it is clear that there exists a time window $[t, t + \varepsilon], t \geq s$ such that $\nu_h^*(\tau) = 1, \tau \in [t, t + \varepsilon]$. We construct another solution as follows:

$$\tilde{\nu}_a(t) = \nu_a^*(t), \tilde{\nu}_h(t) = \nu_h^*(t), \forall t \in [0, s] \text{ and } \tilde{\nu}_a(t) = \tilde{\nu}_h(t) = 0, \forall t \in (s, +\infty],$$

then it is clear that

$$U(\tilde{\nu}_a, \tilde{\nu}_h) - U(\nu_a^*, \nu_h^*) = - \int_s^\infty e^{-rt - \lambda_a \psi_a(t)} ((\lambda_h R - c_h) e^{-\lambda_h \psi_h^*(t)} - \phi_h) \nu_h^*(t) + (\lambda_a R - c_a) \nu_a^*(t) dt > 0,$$

where the inequality holds because $(\lambda_h R - c_h) e^{-\lambda_h \psi_h^*(t)} - \phi_h < 0$ for $t > s$, and there exists a time window $[t, t + \varepsilon]$, $t \geq s$ such that $\nu_h^*(\tau) = 1$, $\tau \in [t, t + \varepsilon]$, and also the fact that $\lambda_a R - c_a \leq 0$. Therefore, under the optimal solution, we must have $\psi_h^*(\infty) \leq \bar{T}_h$.

Given $\psi_h(\infty) \leq \bar{T}_h$, the human integrand is nonnegative, so $e^{-\lambda_a \psi_a(t)} \leq 1$ and $\lambda_a R - c_a \leq 0$ yield

$$U(\nu_h, \nu_a) \leq \int_0^\infty e^{-rt} ((\lambda_h R - c_h) e^{-\lambda_h \psi_h(t)} - \phi_h) \nu_h(t) dt = U(\nu_h, \mathbf{0}),$$

i.e., $\nu_a^* = \mathbf{0}$.

Applying Lemma D13 with $(\rho, \lambda, \delta) = (r, \lambda_h, \lambda_h R - c_h)$ gives $\nu_h^*(t) = \mathbb{1}\{t \leq \bar{T}_h\}$ and

$$\bar{P} = \int_0^{\bar{T}_h} e^{-rt} ((\lambda_h R - c_h) e^{-\lambda_h t} - \phi_h) dt.$$

Step 2: Achievability. Let $T_H := \lambda_h^{-1} \ln((\lambda_h R - c_h)/\phi_h) = \bar{T}_h$. The principal recommends $\nu_h^*(t) = \mathbb{1}\{t \leq T_H\}$ and $\nu_a^* \equiv 0$, terminates if no breakthrough arrives by T_H , and pays

$$L(t) = \frac{\phi_h}{\lambda_h} + \frac{\phi_h}{r} (1 - e^{-r(T_H - t)}).$$

Define $\hat{w}(t) := L(t) - \phi_h/\lambda_h$. A direct computation shows $\hat{w}$ satisfies (2) under (ν_h^*, ν_a^*) with $\hat{w}(T_H) = 0$; hence Lemma D12 gives $\hat{w}(t) = w(t; \Gamma_H^*)$. Since $L(t) = w(t; \Gamma_H^*) + \phi_h/\lambda_h$, the contract is incentive compatible by Lemma 1 (and clearly individually rational), and

$$w(0; \Gamma_H^*) = L(0) - \frac{\phi_h}{\lambda_h} = \frac{\phi_h}{r} (1 - e^{-rT_H}) = \underline{w}(0; \Gamma_H^*). \quad (\text{A.12})$$

Therefore

$$P(\Gamma_H^*) = V(\Gamma_H^*) - w(0; \Gamma_H^*) = (\lambda_h R - c_h) \int_0^{\bar{T}_h} e^{-(r+\lambda_h)t} dt - \phi_h \int_0^{\bar{T}_h} e^{-rt} dt = \bar{P}, \quad (\text{A.13})$$

so Γ_H^* is optimal. Q.E.D.

A.3. Proof of Proposition 2

Throughout, the AI agent is active until success occurs. Following the proof of Lemma 1, for any incentive-compatible contract the human agent's continuation payoff satisfies

$$\dot{w}(t; \Gamma) = (r + \lambda_h \nu_h(t) + \lambda_a) w(t; \Gamma) - \lambda_h \nu_h(t) L(t), \quad (\text{A.14})$$

incentive compatibility is equivalent to

$$\nu_h(t) \left(L(t) - w(t; \Gamma) - \frac{\phi_h}{\lambda_h} \right) \geq 0, \quad (\text{A.15})$$

and individual rationality requires

$$w(t; \Gamma) \geq 0, \quad t \geq 0. \quad (\text{A.16})$$

LEMMA A2 (Human agent promised utility bound). *For any incentive-compatible and individually rational contract Γ , where the principal activates the AI agent until the success occurs,*

$$w(0; \Gamma) \geq \underline{w}(0; \Gamma) := \phi_h \int_0^\infty \nu_h(t) e^{-(r+\lambda_a)t} dt.$$

The proof is similar to that of Lemma A1 and hence omitted.

Step 1: Revenue upper bound. Under any such contract,

$$V(\Gamma) = \int_0^\infty e^{-(r+\lambda_a)t - \lambda_h \psi_h(t)} [(\lambda_a R - c_a) + (\lambda_h R - c_h) \nu_h(t)] dt,$$

so every feasible contract satisfies

$$P(\Gamma) \leq \bar{P} = \max_{\nu_h} U(\nu_h), \text{ with } U(\nu_h) := V(\Gamma) - \underline{w}(0; \Gamma).$$

We first reorganize $U(\nu_h)$, for ease of notations, let

$$Q(t) := e^{-(r+\lambda_a)t - \lambda_h \psi_h(t)} \quad \text{satisfies} \quad \dot{Q}(t) = -(r + \lambda_a + \lambda_h \nu_h(t))Q(t), \text{ with } Q(0) = 0,$$

then

$$\int_0^\infty \dot{Q}(t) dt = -1 = -\int_0^\infty (r + \lambda_h \nu_h(t) + \lambda_a) Q(t) dt,$$

and also

$$\int_0^\infty (\lambda_a R - c_a) Q(t) dt = S_A \int_0^\infty (r + \lambda_a) Q(t) dt = S_A \left(1 - \lambda_h \int_0^\infty \nu_h(t) Q(t) dt \right).$$

Hence

$$U(\nu_h) = S_A + \int_0^\infty \nu_h(t) e^{-(r+\lambda_a)t} \left[(r + \lambda_a + \lambda_h)(S_{AH} - S_A) e^{-\lambda_h \psi_h(t)} - \phi_h \right] dt. \quad (\text{A.17})$$

If $(r + \lambda_a + \lambda_h)(S_{AH} - S_A) \leq \phi_h$, the bracket is nonpositive for all t , so $U \leq S_A$, attained by $\nu_h \equiv 0$; hence $\bar{P} = S_A$. Otherwise, applying Lemma [D13](#) with $(\rho, \lambda, \delta) = (r + \lambda_a, \lambda_h, (r + \lambda_a + \lambda_h)(S_{AH} - S_A))$ gives $\nu_h^*(t) = \mathbb{1}\{t \leq \bar{T}_h\}$ with

$$\bar{T}_h := \frac{1}{\lambda_h} \ln \frac{(r + \lambda_a + \lambda_h)(S_{AH} - S_A)}{\phi_h} = \frac{1}{\lambda_h} \ln \frac{\lambda_h(rR + c_a) - c_h(\lambda_a + r)}{\phi_h(\lambda_a + r)},$$

and

$$\bar{P} = S_A + \int_0^{\bar{T}_h} e^{-(r+\lambda_a)t} \left[(r + \lambda_a + \lambda_h)(S_{AH} - S_A) e^{-\lambda_h t} - \phi_h \right] dt. \quad (\text{A.18})$$

Step 2: Achievability. Let $T_R := \max\{\bar{T}_h, 0\}$. The principal recommends $\nu_h^*(t) = \mathbb{1}\{t \leq T_R\}$, keeps the AI active throughout (also after T_R), and pays

$$L(t) = \frac{\phi_h}{\lambda_h} + \frac{\phi_h}{r + \lambda_a} \left(1 - e^{-(r+\lambda_a)(T_R-t)} \right).$$

With $\hat{w}(t) := L(t) - \phi_h/\lambda_h$, one can easily check that $\hat{w}$ satisfies [\(A.14\)](#) with $\hat{w}(T_R) = 0$, so Lemma [D12](#) (with $\nu_a \equiv 1$) yields $\hat{w}(t) = w(t; \Gamma_R^*)$ and $L(t) = w(t; \Gamma_R^*) + \phi_h/\lambda_h$; feasibility follows from [\(A.15\)](#). In particular,

$$w(0; \Gamma_R^*) = \frac{\phi_h}{r + \lambda_a} \left(1 - e^{-(r+\lambda_a)T_R} \right) = \underline{w}(0; \Gamma_R^*). \quad (\text{A.19})$$

Moreover,

$$P(\Gamma_R^*) = V(\Gamma_R^*) - w(0; \Gamma_R^*) = V(\Gamma_R^*) - \underline{w}(0; \Gamma_R^*) = \bar{P}, \quad (\text{A.20})$$

where the second equality follows from [\(A.19\)](#). Q.E.D.

A.4. Proof of Corollary [1](#)

At the break-even boundary $\lambda_a R = c_a$ we have $S_A = 0$, and by the definitions of T_H and T_R it is clear that $T_H = T_R = \frac{1}{\lambda_h} \ln \frac{\lambda_h R - c_h}{\phi_h} > 0$. Then by [\(A.13\)](#), [\(A.18\)](#) and [\(A.20\)](#),

$$P(\Gamma_H^*) - P(\Gamma_R^*) = \int_0^{T_R} e^{-rt} (1 - e^{-\lambda_a t}) ((\lambda_h R - c_h) e^{-\lambda_h t} - \phi_h) dt > 0,$$

where the inequality holds because $T_R = \frac{1}{\lambda_h} \ln \frac{\lambda_h R - c_h}{\phi_h} > 0$. In addition, by [\(A.12\)](#) and [\(A.19\)](#),

$$w(0; \Gamma_H^*) - w(0; \Gamma_R^*) = \frac{\phi_h}{r} (1 - e^{-rT_H}) - \frac{\phi_h}{r + \lambda_a} (1 - e^{-(r+\lambda_a)T_H}) > 0. \quad Q.E.D.$$

A.5. Proof of Theorem 1

We proceed in three steps: (1) we propose a value function $V(\cdot)$ such that $P(\Gamma) \leq V(w) - w$ for any feasible contract with $w(0; \Gamma) = w$; (2) we derive $\bar{P} := \max_{w \geq 0} \{V(w) - w\}$; (3) we show the proposed contract attains $\bar{P}$.

Step 1. The proposed value function is:

Case 1. If $S_A \geq \max\{S_H, S_{AH}\}$, then

$$V(w) = S_A. \quad (\text{A.21})$$

Case 2. If $S_{AH} > \max\{S_H, S_A\}$, then

$$V(w) = \begin{cases} f_1(w), & 0 \leq w \leq \frac{\phi_h}{\lambda_a + r}, \\ S_{AH}, & w > \frac{\phi_h}{\lambda_a + r}, \end{cases} \quad f_1(w) := S_{AH} + (S_A - S_{AH}) \left(1 - \frac{(\lambda_a + r)w}{\phi_h}\right)^{\frac{\lambda_a + \lambda_h + r}{\lambda_a + r}}. \quad (\text{A.22})$$

Case 3. If $S_H \geq S_{AH}$ and $S_H > S_A$, then

$$V(w) = \begin{cases} f_1(w), & 0 \leq w \leq \eta_0, \\ f_2(w; \eta_0), & \eta_0 < w \leq \frac{\phi_h}{r}, \\ S_H, & w > \frac{\phi_h}{r}, \end{cases} \quad f_2(w; \eta) := S_H + (f_1(\eta) - S_H) \left(1 - \frac{r(w - \eta)}{\phi_h - r\eta}\right)^{\frac{\lambda_h + r}{r}}, \quad (\text{A.23})$$

where $\eta_0 \in [0, \phi_h/(\lambda_a + r)]$ is the unique solution to $f'_1(\eta_0) = \partial f_2(w; \eta_0)/\partial w|_{w=\eta_0}$.

To ensure that the value function in Case 3 is well defined, we first show that η_0 is well defined in the following lemma.

LEMMA A3. Suppose $\lambda_a R > c_a$, $S_H \geq S_{AH}$, and $S_H > S_A$. Then there exists a unique $\eta_0 \in [0, \phi_h/(\lambda_a + r)]$ such that $f'_1(\eta_0) = \frac{\partial f_2(w; \eta_0)}{\partial w}|_{w=\eta_0}$.

Proof: See Appendix D.2 Q.E.D.

To show that for any incentive-compatible contract Γ with $w(0; \Gamma) = w \geq 0$ that $P(\Gamma) \leq V(w) - w$, by the verification Theorem (Theorem 3), it suffices for us to establish the following lemma.

LEMMA A4. The proposed value function V is nonnegative, increasing, and differentiable, and satisfies (D.1) for all $x_a, x_h \in [0, 1]$ and $w \geq 0$.

Proof: See Appendix D.3 Q.E.D.

Step 2. We now determine $\bar{P} := \max_{w \geq 0} \{V(w) - w\}$. We write $V(w; \lambda_a)$ to make the dependence on λ_a explicit (and similarly for other parameters).

LEMMA A5. Define

$$\hat{\lambda}_a := \frac{\lambda_h(rR + c_a)}{c_h + \phi_h} - r, \quad \lambda_a^* := \frac{c_a(r + \lambda_h)}{rR + c_h}, \quad \check{\lambda}_a := \sup \left(\{ \lambda_a \in [\frac{c_a}{R}, \lambda_a^*] : \Delta(\lambda_a) > 0 \} \cup \{ \frac{c_a}{R} \} \right),$$

where

$$\Delta(\lambda_a) := (\lambda_h + r)(S_H - S_{AH}) - \frac{\lambda_a \phi_h}{\lambda_a + r} \left[1 - \frac{\lambda_h}{\lambda_a + \lambda_h + r} K^{\frac{\lambda_a + r}{\lambda_h}} \right], \quad K := \frac{\phi_h}{(\lambda_a + \lambda_h + r)(S_{AH} - S_A)}.$$

Then the unique global maximizer of $V(w) - w$ over $w \geq 0$ is

$$w^* = \begin{cases} \eta_0 + \frac{\phi_h - r\eta_0}{r} \left[1 - \left(\frac{\phi_h - r\eta_0}{(\lambda_h + r)(S_H - f_1(\eta_0))} \right)^{\frac{r}{\lambda_h}} \right], & \lambda_a < \check{\lambda}_a, \\ \frac{\phi_h}{\lambda_a + r} \left(1 - K^{\frac{\lambda_a + r}{\lambda_h}} \right), & \check{\lambda}_a \leq \lambda_a < \hat{\lambda}_a, \\ 0, & \lambda_a \geq \hat{\lambda}_a, \end{cases} \quad (\text{A.24})$$

³ With slight abuse of notation, we write w for the agent's total expected utility at time 0 under contract Γ . We adopt the same convention whenever no confusion arises.

and

$$\bar{P} = \begin{cases} S_H + (f_1(\eta_0) - S_H) \left(\frac{\phi_h - r\eta_0}{(\lambda_h + r)(S_H - f_1(\eta_0))} \right)^{\frac{\lambda_h + r}{\lambda_h}} - w^*, & \lambda_a < \check{\lambda}_a, \\ S_{AH} - \frac{\phi_h}{\lambda_a + \lambda_h + r} K^{\frac{\lambda_a + r}{\lambda_h}} - \frac{\phi_h}{\lambda_a + r} \left(1 - K^{\frac{\lambda_a + r}{\lambda_h}} \right), & \check{\lambda}_a \leq \lambda_a < \hat{\lambda}_a, \\ S_A, & \lambda_a \geq \hat{\lambda}_a. \end{cases} \quad (\text{A.25})$$

Proof: See Appendix [D.4](#) Q.E.D.

Step 3. Finally, we show that, the proposed contract in Theorem [1](#) is feasible and attains the principal's maximal payoff. The proposed contract is $\Gamma_{HA}(T_1^*, T_2^*)$, where:

DEFINITION 2 (CONTRACT $\Gamma_{HA}(T_1, T_2)$). Given switching dates $0 \leq T_1 \leq T_2 \leq \infty$:

- Stage 1 ($t \in [0, T_1]$): only the human agent is active; a breakthrough at t pays $L(t) = e^{-r(T_1-t)} \frac{\phi_h}{r+\lambda_a} (1 - e^{-(r+\lambda_a)(T_2-T_1)}) + \frac{\phi_h}{r} (1 - e^{-r(T_1-t)}) + \frac{\phi_h}{\lambda_h}$.
- Stage 2 ($t \in (T_1, T_2]$): both agents are active; a human breakthrough at t pays $L(t) = \frac{\phi_h}{r+\lambda_a} (1 - e^{-(r+\lambda_a)(T_2-t)}) + \frac{\phi_h}{\lambda_h}$.
- Stage 3 ($t \in (T_2, \infty)$): only the AI agent is active.

If the breakthrough is generated by the AI, or there is no breakthrough, the human agent receives no payment. The contract terminates upon breakthrough. $\square$

LEMMA A6. Contract $\Gamma_{HA}(T_1, T_2)$ is incentive compatible. The agent's payoff-to-go is

$$w(t) = \begin{cases} e^{-r(T_1-t)} \frac{\phi_h}{r+\lambda_a} (1 - e^{-(r+\lambda_a)(T_2-T_1)}) + \frac{\phi_h}{r} (1 - e^{-r(T_1-t)}), & t \in [0, T_1], \\ \frac{\phi_h}{r+\lambda_a} (1 - e^{-(r+\lambda_a)(T_2-t)}), & t \in (T_1, T_2], \\ 0, & t \in (T_2, \infty), \end{cases}$$

and the principal's total expected payoff is

$$P(\Gamma_{HA}(T_1, T_2)) = S_H (1 - e^{-(\lambda_h+r)T_1}) + e^{-(\lambda_h+r)T_1} [S_{AH} + (S_A - S_{AH})e^{-(\lambda_a+\lambda_h+r)(T_2-T_1)}] - w(0).$$

Proof: See Appendix [D.5](#) Q.E.D.

We check that $\Gamma_{HA}(T_1^*, T_2^*)$ attains $\bar{P}$ in each of the three regions.

If $\lambda_a \geq \hat{\lambda}_a$: $T_1^* = T_2^* = 0$, so $w(0) = 0$ and $P(\Gamma_{HA}(0, 0)) = S_A = \bar{P}$.

If $\check{\lambda}_a \leq \lambda_a < \hat{\lambda}_a$: $T_1^* = 0$ and $T_2^* = \lambda_h^{-1} \ln \left(\frac{\lambda_h(rR+c_a) - (\lambda_a+r)c_h}{\phi_h(\lambda_a+r)} \right)$, so $e^{-\lambda_h T_2^*} = K$, and hence $e^{-(\lambda_a+r)T_2^*} = K^{\frac{\lambda_a+r}{\lambda_h}}$ and $e^{-(\lambda_a+\lambda_h+r)T_2^*} = K^{\frac{\lambda_a+\lambda_h+r}{\lambda_h}}$. Then $w(0) = \frac{\phi_h}{\lambda_a+r} (1 - K^{\frac{\lambda_a+r}{\lambda_h}}) = w^*$, and substituting into Lemma [A6](#),

$$P(\Gamma_{HA}(T_1^*, T_2^*)) = S_{AH} + (S_A - S_{AH})K^{\frac{\lambda_a+\lambda_h+r}{\lambda_h}} - \frac{\phi_h}{\lambda_a+r} \left(1 - K^{\frac{\lambda_a+r}{\lambda_h}} \right) = f_1(w^*) - w^* = \bar{P}.$$

If $\lambda_a < \check{\lambda}_a$:

$$T_1^* = \frac{1}{\lambda_h} \ln \left(\frac{(\lambda_h+r)(S_H - f_1(\eta_0))}{\phi_h - r\eta_0} \right), \quad T_2^* = \frac{1}{\lambda_h} \ln \left(\frac{\lambda_h(rR+c_a) - c_h(r+\lambda_a)}{\phi_h(r+\lambda_a)} \right).$$

Taking logarithms in the smooth-pasting condition $f_1'(\eta_0) = \partial_w f_2(w; \eta_0)|_{w=\eta_0}$, written as

$$\frac{\lambda_h(rR+c_a) - c_h(r+\lambda_a)}{\phi_h(r+\lambda_a)} \left(1 - \frac{(\lambda_a+r)\eta_0}{\phi_h} \right)^{\frac{\lambda_h}{\lambda_a+r}} = \frac{(\lambda_h+r)(S_H - f_1(\eta_0))}{\phi_h - r\eta_0},$$

gives

$$T_2^* - T_1^* = \frac{1}{\lambda_a+r} \ln \left(\frac{\phi_h}{\phi_h - (\lambda_a+r)\eta_0} \right), \quad \text{i.e.,} \quad e^{-(\lambda_a+r)(T_2^* - T_1^*)} = \frac{\phi_h - (\lambda_a+r)\eta_0}{\phi_h}.$$

Also $e^{-rT_1^*} = \left(\frac{\phi_h - r\eta_0}{(\lambda_h+r)(S_H - f_1(\eta_0))} \right)^{r/\lambda_h}$, so

$$w(0) = e^{-rT_1^*} \frac{\phi_h}{\lambda_a+r} \left(1 - e^{-(\lambda_a+r)(T_2^* - T_1^*)} \right) + \frac{\phi_h}{r} (1 - e^{-rT_1^*}) = e^{-rT_1^*} \eta_0 + \frac{\phi_h}{r} (1 - e^{-rT_1^*}) = w^*.$$

Substituting into Lemma [A6](#) and using $S_{AH} + (S_A - S_{AH})e^{-(\lambda_a+\lambda_h+r)(T_2^* - T_1^*)} = f_1(\eta_0)$,

$$P(\Gamma_{HA}(T_1^*, T_2^*)) = S_H (1 - e^{-(\lambda_h+r)T_1^*}) + e^{-(\lambda_h+r)T_1^*} f_1(\eta_0) - w(0) = f_2(w^*; \eta_0) - w^* = \bar{P}.$$

In all three cases $P(\Gamma_{HA}(T_1^*, T_2^*)) = \bar{P}$, so the proposed contract is optimal. Q.E.D.

Appendix B: Proofs for Section 5

B.1. Proof of Proposition 3

To keep notation clean we drop the superscript HH . Following the proof of Lemma 1 for any incentive-compatible contract the continuation payoff of agent $i \in \{1, 2\}$ satisfies

$$\dot{w}_i(t; \Gamma) = (r + \lambda_h(\nu_1(t) + \nu_2(t)))w_i(t; \Gamma) - \lambda_h \nu_i(t) L_i(t), \quad (\text{B.1})$$

incentive compatibility is equivalent to

$$\nu_i(t) \left(L_i(t) - w_i(t; \Gamma) - \frac{\phi_h}{\lambda_h} \right) \geq 0, \quad (\text{B.2})$$

and individual rationality requires

$$w_i(t; \Gamma) \geq 0, \quad t \geq 0. \quad (\text{B.3})$$

Let $H(t) := \psi_1(t) + \psi_2(t)$, where $\psi_i(t) := \int_0^t \nu_i(s) ds$.

LEMMA B7 (Promised utility lower bound). *For any incentive-compatible and individually rational contract Γ ,*

$$w_i(0; \Gamma) \geq \phi_h \int_0^\infty \nu_i(t) e^{-rt - \lambda_h H(t) + \lambda_h \psi_i(t)} dt, \quad i = 1, 2, \quad (\text{B.4})$$

and

$$w_1(0; \Gamma) + w_2(0; \Gamma) \geq \phi_h \int_0^\infty e^{-rt - \lambda_h H(t)/2} dH(t). \quad (\text{B.5})$$

Proof: See Appendix D.6 Q.E.D.

Step 1: Revenue upper bound. For any contract Γ , the societal value is $\int_0^\infty e^{-rt - \lambda_h H(t)} (\lambda_h R - c_h) dH(t)$, so by (B.5),

$$P(\Gamma) \leq \int_0^\infty e^{-rt - \lambda_h H(t)/2} \left[(\lambda_h R - c_h) e^{-\lambda_h H(t)/2} - \phi_h \right] dH(t). \quad (\text{B.6})$$

Define $\hat{H} := \frac{2}{\lambda_h} \ln \left(\frac{\lambda_h R - c_h}{\phi_h} \right)$; the bracket is nonnegative iff $H(t) \leq \hat{H}$. Since at most two agents work at any time, $H(t) \leq 2t$, so $e^{-rt} \leq e^{-rH(t)/2}$. Dropping nonpositive marginal contributions and changing variables to $h = H(t)$,

$$P(\Gamma) \leq \bar{P} := \int_0^{\hat{H}} \left[(\lambda_h R - c_h) e^{-(r+2\lambda_h)h/2} - \phi_h e^{-(r+\lambda_h)h/2} \right] dh. \quad (\text{B.7})$$

Step 2: Achievability. Let $T_{HH} := \lambda_h^{-1} \ln \left(\frac{\lambda_h R - c_h}{\phi_h} \right)$. The principal recommends $\nu_1(t) = \nu_2(t) = \mathbb{1}\{t \leq T_{HH}\}$, terminates if no breakthrough arrives by T_{HH} , and pays each agent

$$L_i(t) = \frac{\phi_h}{\lambda_h} + \frac{\phi_h}{\lambda_h + r} \left(1 - e^{-(\lambda_h + r)(T_{HH} - t)} \right), \quad 0 \leq t \leq T_{HH}.$$

With $\hat{w}_i(t) := L_i(t) - \phi_h/\lambda_h$, one verifies the promise-keeping ODE (B.1) with $\hat{w}_i(T_{HH}) = 0$, so by Lemma D12 (replacing λ_a, ν_a, ν_h with λ_h, ν_1, ν_2), $\hat{w}_i(t) = w_i(t; \Gamma_{HH}^*)$. Hence $L_i(t) = w_i(t; \Gamma_{HH}^*) + \phi_h/\lambda_h$, feasibility follows from (B.2), and

$$w_i(0; \Gamma_{HH}^*) = L_i(0) - \frac{\phi_h}{\lambda_h} = \frac{\phi_h}{\lambda_h + r} \left[1 - \left(\frac{\phi_h}{\lambda_h R - c_h} \right)^{\frac{\lambda_h + r}{\lambda_h}} \right]. \quad (\text{B.8})$$

Finally, $H(t) = 2t$ on $[0, T_{HH}]$, so by the change of variables $h = 2t$,

$$P(\Gamma_{HH}^*) = \int_0^{\hat{H}} e^{-(r+2\lambda_h)h/2} (\lambda_h R - c_h) dh - \frac{2\phi_h}{\lambda_h + r} \left(1 - e^{-(\lambda_h + r)\hat{H}/2} \right) = \bar{P},$$

so Γ_{HH}^* attains the upper bound and is optimal. Q.E.D.

B.2. Proof of Proposition 4

We write $P(\Gamma_{HA}^*(\lambda_a))$ and $w^*(0; \Gamma_{HA}^*(\lambda_a))$ for the principal's and agent's payoffs under the optimal contract as functions of λ_a (similarly for c_a). Note S_H does not depend on λ_a , and

$$S'_A(\lambda_a) = \frac{rR + c_a}{(\lambda_a + r)^2} > 0, \quad S'_{AH}(\lambda_a) = \frac{rR + c_a + c_h}{(\lambda_a + \lambda_h + r)^2} > 0. \quad (\text{B.9})$$

Now we turn to summarize some useful properties of $f_1(w; \lambda_a)$ and $f_2(w; \lambda_a)$ in the following lemma.

LEMMA B8. *When $w \in [0, \frac{\phi_h}{\lambda_a + r}]$ and $S_{AH} > \max\{S_A, S_H\}$: for any given w , $\partial f_1(w; \lambda_a)/\partial w$ is decreasing in λ_a , $\partial f_1(w; c_a)/\partial w$ is increasing in c_a , and $f_1(w; \lambda_a)$ is increasing in λ_a . When $w \in [\eta_0, \frac{\phi_h}{r}]$ and $S_H \geq S_{AH}$, $S_H > S_A$: for any given w , $\partial f_2(w; \lambda_a)/\partial w$ is decreasing in λ_a , $\partial f_2(w; c_a)/\partial w$ is increasing in c_a , and $f_2(w; \lambda_a)$ is increasing in λ_a .*

Proof: See Appendix D.7. Q.E.D.

We claim $P(\Gamma_{HA}^*(\lambda_a))$ is increasing in λ_a , $w^*(0; \Gamma_{HA}^*(\lambda_a))$ is decreasing in λ_a , and $w^*(0; \Gamma_{HA}^*(c_a))$ is increasing in c_a by distinguishing among three cases.

- (a) If $S_A \geq \max\{S_H, S_{AH}\}$: $P(\Gamma_{HA}^*(\lambda_a)) = S_A(\lambda_a)$, strictly increasing by (B.9), and $w^*(0; \Gamma_{HA}^*(\lambda_a)) = w^*(0; \Gamma_{HA}^*(c_a)) = 0$, which is clearly decreasing in λ_a and increasing in c_a .
- (b) If $S_{AH} > \max\{S_H, S_A\}$: by definition,

$$P(\Gamma_{HA}^*(\lambda_a)) = \max_{w \geq 0} \{V(w; \lambda_a) - w\}, \quad w^*(\lambda_a) := w^*(0; \Gamma_{HA}^*(\lambda_a)) = \arg \max_{w \geq 0} \{V(w; \lambda_a) - w\},$$

$$\text{and } w^*(0; \Gamma_{HA}^*(c_a)) = \arg \max_{w \geq 0} \{V(w; c_a) - w\}.$$

where

$$V(w; \lambda_a) = \begin{cases} f_1(w; \lambda_a), & 0 \leq w \leq \frac{\phi_h}{\lambda_a + r}, \\ S_{AH}(\lambda_a), & w > \frac{\phi_h}{\lambda_a + r}. \end{cases}$$

which is clearly increasing in λ_a by (B.9) and Lemma B8. Therefore $P(\Gamma_{HA}^*(\lambda_a))$ is also increasing in λ_a . Since $w^*(\lambda_a) \in [0, \frac{\phi_h}{\lambda_a + r})$, by FOC, we have $\frac{\partial f_1(w; \lambda_a)}{\partial w}|_{w=w^*(\lambda_a)} = 1$. Taking derivative over λ_a on both sides, we know that

$$\frac{\partial^2 f_1(w; \lambda_a)}{\partial^2 w}|_{w=w^*(\lambda_a)} \frac{dw^*(\lambda_a)}{d\lambda_a} + \frac{\partial^2 f_1(w; \lambda_a)}{\partial \lambda_a \partial w}|_{w=w^*(\lambda_a)} = 0,$$

Since $\frac{\partial^2 f_1(w; \lambda_a)}{\partial w \partial \lambda_a} \leq 0$ for $w \in [0, \frac{\phi_h}{\lambda_a + r}]$ by Lemma B8, and $\frac{\partial^2 f_1(w; \lambda_a)}{\partial^2 w}|_{w=w^*(\lambda_a)} < 0$, it is clear $\frac{dw^*(\lambda_a)}{d\lambda_a} \leq 0$, i.e., $w^*(\lambda_a)$ is decreasing in λ_a . By similar argument, since $\frac{\partial^2 f_1(w; c_a)}{\partial w \partial c_a} \geq 0$ for $w \in [0, \frac{\phi_h}{\lambda_a + r}]$, we can show that $w^*(0; \Gamma_{HA}^*(c_a))$ is increasing in c_a .

- (c) If $S_H \geq S_{AH}$ and $S_H > S_A$: V takes the three-piece form (A.23), increasing in λ_a by Lemma B8, hence so is $P(\Gamma_{HA}^*(\lambda_a))$. If $w^*(\lambda_a) \leq \eta_0$, case (b) applies (since $\eta_0 < \frac{\phi_h}{\lambda_a + r}$); if $w^*(\lambda_a) \in (\eta_0, \frac{\phi_h}{r}]$, the identical FOC argument applied to f_2 (using $\frac{\partial^2 f_2(w; \lambda_a)}{\partial w \partial \lambda_a} \leq 0$ and $\frac{\partial^2 f_2(w; \lambda_a)}{\partial^2 w}|_{w=w^*(\lambda_a)} < 0$, $\frac{\partial^2 f_2(w; c_a)}{\partial w \partial c_a} \geq 0$) gives $w^*(0; \Gamma_{HA}^*(\lambda_a))$ is decreasing in λ_a , and $w^*(0; \Gamma_{HA}^*(c_a))$ is increasing in c_a .

Since $P(\Gamma_{HA}^*(\lambda_a))$, $w^*(0; \Gamma_{HA}^*(\lambda_a))$, and $w^*(0; \Gamma_{HA}^*(c_a))$ are continuous, monotonicity on each piece implies global monotonicity. Q.E.D.

B.3. Proof of Proposition 5

Since $P(\Gamma_{HA}^*(\lambda_a; c_a))$ is increasing in λ_a , by the definition of λ_a^P in (4),

$$\lambda_a^P(c_a) \leq \lambda_a^A(c_a) \iff Q(c_a) := P(\Gamma_{HA}^*(\lambda_a^A(c_a)); c_a) \geq P_{HH}^*. \quad (\text{B.10})$$

To show there exists c_a^* such that $[0, c_a^*] = \{c_a : Q(c_a) \geq P_{HH}^*\}$, we first present the following lemma.

LEMMA B9. *Given $c_a \in [0, \infty)$: if $\arg \max_{w \geq 0} \{V(w; \lambda_a^A(c_a), c_a) - w\} = \arg \max_{w \geq 0} \{f_2(w; \lambda_a^A(c_a), c_a) - w\}$, then $Q(c_a) < P_{HH}^*$. If instead $\arg \max_{w \geq 0} \{V(w; \lambda_a^A(c_a), c_a) - w\} = \arg \max_{w \geq 0} \{f_1(w; \lambda_a^A(c_a), c_a) - w\}$, then $Q(c_a)$ is decreasing in c_a .*

Proof: See Appendix D.8. Q.E.D.

By Lemma B9 and continuity of Q , there exists $c_a^* = \sup\{c_a \in [0, \infty) : Q(c_a) \geq P_{HH}^*\}$ such that $Q(c_a) \geq P_{HH}^*$ iff $c_a \leq c_a^*$. Q.E.D.

B.4. Proof of Corollary 3

With the definition of λ_a^P , and λ_a^A , by Proposition 4, $P(\Gamma_{HA}^*(\lambda_a)) \geq P_{HH}$ iff $\lambda_a \geq \lambda_a^P$, and $w^*(0; \Gamma_{HA}^*(\lambda_a)) \geq w_{HH}^*$ iff $\lambda_a \leq \lambda_a^A$. Hence, if $\lambda_a \leq \underline{\lambda}_a := \min\{\lambda_a^P, \lambda_a^A\}$, then $P(\Gamma_{HA}^*(\lambda_a)) \leq P_{HH}$ and $w^*(0; \Gamma_{HA}^*(\lambda_a)) \geq w_{HH}^*$; if $\lambda_a > \bar{\lambda}_a := \max\{\lambda_a^P, \lambda_a^A\}$, both inequalities reverse. If $\underline{\lambda}_a < \lambda_a \leq \bar{\lambda}_a$, by Proposition 5, $\lambda_a^P \leq \lambda_a^A$ if and only if $c_a \leq c_a^*$. Again by the definition of λ_a^P , and λ_a^A , as well as Proposition 4, we have if $c_a \leq c_a^*$, then when $\underline{\lambda}_a = \lambda_a^P < \lambda_a \leq \bar{\lambda}_a = \lambda_a^A$, $P(\Gamma_{HA}^*(\lambda_a)) \geq P_{HH}$ and $w^*(0; \Gamma_{HA}^*(\lambda_a)) \geq w_{HH}^*$; and if $c_a > c_a^*$, then when $\underline{\lambda}_a = \lambda_a^A < \lambda_a \leq \bar{\lambda}_a = \lambda_a^P$, $P(\Gamma_{HA}^*(\lambda_a)) \leq P_{HH}$ and $w^*(0; \Gamma_{HA}^*(\lambda_a)) \leq w_{HH}^*$; Q.E.D.

Appendix C: Proofs in Section 6

C.1. Proof of Theorem 2

Following the proof of Lemma 1, for any incentive-compatible contract the human agent's continuation payoff satisfies

$$\dot{w}(t; \Gamma) = (r + \lambda_h \nu_h(t) + \lambda_a \nu_a(t)(1 - \kappa(t)) + p\lambda_a \nu_a(t)\kappa(t))w(t; \Gamma) - \lambda_h \nu_h(t)L(t), \quad (C.1)$$

incentive compatibility is equivalent to

$$\nu_h(t) \left(L(t) - w(t; \Gamma) - \frac{\phi_h}{\lambda_h} \right) \geq 0, \quad (C.2)$$

and individual rationality requires

$$w(t; \Gamma) \geq 0, \quad t \geq 0. \quad (C.3)$$

Define the AI-mode indicators

$$x^N(t) := 1 - \nu_a(t), \quad x^{AC}(t) := \nu_a(t)\kappa(t), \quad x^A(t) := \nu_a(t)(1 - \kappa(t)),$$

so $x^N(t) + x^{AC}(t) + x^A(t) = 1$. Here, $x^N(t) = 1$ means that AI is not used, $x^{AC}(t) = 1$ means that AI is active and inspected/checked, and $x^A(t) = 1$ means that AI is active and uninspected. Set

$$\psi_a(t) := \int_0^t (x^A(s) + px^{AC}(s))ds, \quad Q(t) := e^{-rt - \lambda_h \psi_h(t) - \lambda_a \psi_a(t)}, \quad \tilde{S}_A := \frac{p\lambda_a R - c_a}{r + \lambda_a} \geq 0.$$

LEMMA C10 (Human agent promised utility lower bound). For any incentive-compatible and individually rational contract Γ ,

$$w(0; \Gamma) \geq \underline{w}(0; \Gamma) := \phi_h \int_0^\infty \nu_h(t) e^{-rt - \lambda_a \psi_a(t)} dt.$$

The proof is similar to that of Lemma A1 and hence omitted.

Step 1: Revenue upper bound. Under any contract, the societal value is

$$V(\Gamma) = \int_0^\infty Q(t) \left[(\lambda_h R - c_h) \nu_h(t) + (p\lambda_a R - c_a) x^A(t) + (p\lambda_a R - c_a - \lambda_a K) x^{AC}(t) \right] dt,$$

so, by Lemma C10, $P(\Gamma) \leq U(\nu_h, \mathbf{x}^N, \mathbf{x}^{AC}, \mathbf{x}^A) := V(\Gamma) - \phi_h \int_0^\infty \nu_h(t) e^{-rt - \lambda_a \psi_a(t)} dt$. Proceeding exactly as in Step 1 of the proof of Proposition 2 (using $\int_0^\infty \dot{Q}(t) dt = -1$ and the definition of $\tilde{S}_A$), a direct computation gives

$$U(\nu_h, \mathbf{x}^N, \mathbf{x}^{AC}, \mathbf{x}^A) = \tilde{S}_A + \int_0^\infty e^{-rt - \lambda_a \psi_a(t)} (\delta e^{-\lambda_h \psi_h(t)} - \phi_h) \nu_h(t) dt - \int_0^\infty Q(t) (r\tilde{S}_A x^N(t) + \eta x^{AC}(t)) dt,$$

where $\delta := \lambda_h R - c_h - \lambda_h \tilde{S}_A$ and $\eta := (r + p\lambda_a) \tilde{S}_A - (p\lambda_a R - c_a - \lambda_a K) = \lambda_a (K - K^*) \geq 0$, using $K^* = (1 - p)\tilde{S}_A$.

In the following, we solve

$$\bar{P} := \max_{\nu_h(t) \in \{0,1\}, x^N(t) + x^{AC}(t) + x^A(t) = 1, x^N(t) \geq 0, \forall t} U(\nu_h, \mathbf{x}^N, \mathbf{x}^{AC}, \mathbf{x}^A). \quad (C.4)$$

Let $\bar{T}_h := \lambda_h^{-1} \ln(\max\{\delta/\phi_h, 1\})$. As in the proof of Lemma D13, it is without loss to restrict to $\psi_h(\infty) \leq \bar{T}_h$. Fixing the AI controls and letting $G(x) := \int_0^x (\delta e^{-\lambda_h s} - \phi_h) ds$, integration by parts, $\psi_h(t) \leq \min\{t, \bar{T}_h\}$, and monotonicity of G on $[0, \bar{T}_h]$ give

$$U(\nu_h, \mathbf{x}^N, \mathbf{x}^{AC}, \mathbf{x}^A) \leq \tilde{S}_A + \int_0^{\bar{T}_h} e^{-rt - \lambda_a \psi_a(t)} (\delta e^{-\lambda_h t} - \phi_h) dt - \int_0^{\bar{T}_h} e^{-rt - \lambda_h t - \lambda_a \psi_a(t)} (r\tilde{S}_A x^N(t) + \eta x^{AC}(t)) dt,$$

attained by $\nu_h^*(t) = \mathbb{1}\{t \leq \bar{T}_h\}$ and, since the post- $\bar{T}_h$ penalty terms are nonpositive, $x^{A*}(t) = 1$ for $t \geq \bar{T}_h$. The remaining choice of AI mode on $[0, \bar{T}_h]$ solves

$$\begin{aligned} \max_{x^N, x^{AC}, x^A} & \left\{ \tilde{S}_A + \int_0^{\bar{T}_h} e^{-rt - \lambda_a \psi_a(t)} (\delta e^{-\lambda_h t} - \phi_h) dt - \int_0^{\bar{T}_h} e^{-rt - \lambda_h t - \lambda_a \psi_a(t)} (r \tilde{S}_A x^N + \eta x^{AC}) dt \right\} \\ \text{s.t.} & \quad x^N(t) + x^{AC}(t) + x^A(t) = 1, \quad x^N, x^{AC}, x^A \geq 0, \quad \forall t \in [0, \bar{T}_h]. \end{aligned} \quad (\mathcal{P}_a)$$

LEMMA C11. *The optimal solution of (D.13) is as follows: there exist switch times $0 \leq T_1^* \leq T_2^* \leq T_3^*$ such that*

$$x^{N*}(t) = 1 \text{ on } [0, T_1^*), \quad x^{AC*}(t) = 1 \text{ on } [T_1^*, T_2^*), \quad x^{A*}(t) = 1 \text{ on } [T_2^*, \infty),$$

together with $\nu_h^*(t) = \mathbf{1}\{t < \bar{T}_h\}$.

Proof: See Appendix D.9. Q.E.D.

Step 2: Achievability. Fix $0 \leq T_1 \leq T_2 \leq T_3$ and consider $\Gamma_{H-AC-A}(T_1, T_2, T_3)$ with payment $L(t)$ as specified in the theorem; define $\hat{w}(t) := L(t) - \phi_h/\lambda_h$. Under this contract,

$$\nu_h(t) = \mathbb{1}\{t \leq T_3\}, \quad x^N(t) = \mathbb{1}\{t < T_1\}, \quad x^{AC}(t) = \mathbb{1}\{T_1 \leq t < T_2\}, \quad x^A(t) = \mathbb{1}\{t \geq T_2\},$$

and one verifies directly that $\hat{w}$ satisfies (C.1) with $\hat{w}(T_3) = 0$. Hence, by Lemma D12 (replacing $\nu_a(t)$ with $px^{AC}(t) + x^A(t)$), $\hat{w}(t) = w(t; \Gamma)$, so $L(t) = w(t; \Gamma) + \phi_h/\lambda_h$ and the contract is incentive compatible by (C.2), and clearly individually rational.

Finally, let $\Gamma_{HAC}^* := \Gamma_{H-AC-A}(T_1^*, T_2^*, \bar{T}_h)$. Since $L^*(t) = w(t; \Gamma_{HAC}^*) + \phi_h/\lambda_h$ binds,

$$\frac{dw(t; \Gamma_{HAC}^*)}{dt} = (r + \lambda_a x^{A*}(t) + p \lambda_a x^{AC*}(t)) w(t; \Gamma_{HAC}^*) - \phi_h \nu_h^*(t),$$

which integrates to $w(0; \Gamma_{HAC}^*) = \phi_h \int_0^{\bar{T}_h} e^{-rt - \lambda_a \psi_a(t)} dt = \underline{w}(0; \Gamma_{HAC}^*)$, the lower bound of Lemma C10 evaluated at the optimal working scheme. Since Γ_{HAC}^* implements exactly the working scheme that attains the maximum of (C.4), and the societal value depends only on the working scheme,

$$P(\Gamma_{HAC}^*) = V^* - w(0; \Gamma_{HAC}^*) = V^* - \underline{w}(0; \Gamma_{HAC}^*) = \bar{P},$$

so the proposed mechanism is optimal. Q.E.D.

Appendix D: Auxiliary Lemmas

LEMMA D12. *Suppose that under contract Γ the human agent is paid $L(\tau)$ only when he generates the success, and that $\hat{w}(\cdot)$ evolves according to (2), i.e.,*

$$\frac{d\hat{w}(t)}{dt} = (\lambda_a \nu_a(t) + \lambda_h \nu_h(t) + r) \hat{w}(t) - \lambda_h \nu_h(t) L(t),$$

with $\hat{w}(\infty) = 0$. Then the human agent's expected payoff-to-go at t is

$$w(t; \Gamma) := \mathbb{E}[e^{-r(\tau_h - t)} L(\tau_h) \mathbb{1}(\tau_h < \tau_a) \mid \tau \geq t] = \hat{w}(t).$$

Proof: Solving the linear ODE with terminal condition $\hat{w}(\infty) = 0$,

$$\hat{w}(t) = \int_t^\infty e^{-\int_t^\tau (\lambda_a \nu_a(s) + \lambda_h \nu_h(s) + r) ds} \lambda_h \nu_h(\tau) L(\tau) d\tau = \mathbb{E}[e^{-r(\tau_h - t)} L(\tau_h) \mathbb{1}(\tau_h < \tau_a) \mid \tau \geq t],$$

since the survival rate is $e^{-\int_t^\tau (\lambda_a \nu_a + \lambda_h \nu_h) ds}$ and the human agent succeeds at rate $\lambda_h \nu_h(\tau)$. Q.E.D.

THEOREM 3 (Verification Theorem). *Suppose there exists a nondecreasing, differentiable function $V : \mathbb{R} \rightarrow \mathbb{R}_+$ such that for any $x_a, x_h \in \{0, 1\}$ and any $\hat{w} \in \mathbb{R}_+$,*

$$-(c_a x_a + c_h x_h) + (\lambda_a x_a + \lambda_h x_h) R + ((r + \lambda_a x_a) \hat{w} - \phi_h x_h) V'(\hat{w}) - (r + \lambda_a x_a + \lambda_h x_h) V(\hat{w}) \leq 0. \quad (\text{D.1})$$

Then, for any feasible contract Γ with $w(0; \Gamma) = \hat{w}$, $P(\Gamma) \leq V(\hat{w}) - \hat{w}$.

Proof: Let $m(t) := \lambda_a \nu_a(t) + \lambda_h \nu_h(t)$ and $g(t) := e^{-\int_0^t m(s) ds - rt} V(w(t))$. Then $\dot{g}(t) = e^{-\int_0^t m(s) ds - rt} (\dot{w}(t) V'(w(t)) - (m(t) + r) V(w(t)))$, and note that

$$\begin{aligned} \dot{w}(t) V'(w(t)) - (m(t) + r) V(w(t)) &= ((\lambda_a \nu_a(t) + \lambda_h \nu_h(t) + r) w(t) - \lambda_h \nu_h(t) L(t)) V'(w(t)) - (m(t) + r) V(w(t)) \\ &\leq ((\lambda_a \nu_a(t) + r) w(t) - \phi_h \nu_h(t)) V'(w(t)) - (m(t) + r) V(w(t)) \\ &\leq -(\lambda_a \nu_a(t) + \lambda_h \nu_h(t)) R + (c_a \nu_a(t) + c_h \nu_h(t)), \end{aligned}$$

where the second equality comes from (2), the first inequality holds due to (IC(t)) and $V'(\cdot) \geq 0$, and the last inequality uses (D.1) with $x_i, i \in \{a, h\}$ replaced by $\nu_i(t)$. Therefore, $\dot{g}(t) \leq -e^{-\int_0^t m(s) ds - rt} (m(t) R - (c_a \nu_a(t) + c_h \nu_h(t)))$. Integrating and using $g(\infty) \geq 0$,

$$V(\hat{w}) = g(0) \geq \int_0^\infty e^{-\int_0^t m(s) ds - rt} (m(t) R - (c_a \nu_a(t) + c_h \nu_h(t))) dt = \mathbb{E} \left[e^{-r\tau} R \mathbb{1}(\tau < \infty) - \int_0^\tau e^{-rt} (c_a \nu_a + c_h \nu_h) dt \right], \quad (\text{D.2})$$

where the last equality uses (D.3) below. Under contract Γ , with $w(0; \Gamma) = \hat{w}$, the societal value is:

$$P(\Gamma) + w(0; \Gamma) = P(\Gamma) + \hat{w} = \mathbb{E} \left[e^{-r\tau} R \mathbb{1}(\tau < \infty) - \int_0^\tau e^{-rt} (c_a \nu_a(t) + c_h \nu_h(t)) dt \right] \leq V(\hat{w}),$$

where the inequality follows from (D.2). This yields the desired result.

For future reference, we record the societal-value decomposition: with $m(t)$ as above,

$$\mathbb{E} \left[e^{-r\tau} R \mathbb{1}(\tau < \infty) - \int_0^\tau e^{-rt} (c_a \nu_a(t) + c_h \nu_h(t)) dt \right] = \int_0^\infty e^{-\int_0^t m(s) ds - rt} (m(t) R - (c_a \nu_a(t) + c_h \nu_h(t))) dt, \quad (\text{D.3})$$

which follows from $\mathbb{E}[e^{-r\tau} R \mathbb{1}(\tau < \infty)] = \int_0^\infty e^{-\int_0^t m(s) ds - rt} m(t) R dt$ and $\mathbb{P}(t < \tau) = e^{-\int_0^t m(s) ds}$. Q.E.D.

LEMMA D13 (Front-loading). Fix $\rho > 0$, $\lambda > 0$, $\phi_h > 0$, and $\delta \in \mathbb{R}$, and consider

$$\max_{\nu(t) \in \{0,1\}, \forall t} \int_0^\infty e^{-\rho t} (\delta e^{-\lambda \psi(t)} - \phi_h) \nu(t) dt, \quad \psi(t) := \int_0^t \nu(s) ds.$$

Let $\bar{T} := \lambda^{-1} \ln(\delta/\phi_h)$ if $\delta > \phi_h$ and $\bar{T} := 0$ otherwise. Then $\nu^*(t) = \mathbb{1}\{t \leq \bar{T}\}$ is optimal, with optimal value $\int_0^{\bar{T}} e^{-\rho t} (\delta e^{-\lambda t} - \phi_h) dt$.

Proof: The marginal value of effort at cumulative exposure x is $\delta e^{-\lambda x} - \phi_h$, which is negative for $x > \bar{T}$; hence it is without loss to restrict to $\psi(\infty) \leq \bar{T}$ (stopping all effort once exposure $\bar{T}$ is reached weakly improves the objective). Define $G(x) := \int_0^x (\delta e^{-\lambda s} - \phi_h) ds$, increasing on $[0, \bar{T}]$. Since $\frac{d}{dt} G(\psi(t)) = (\delta e^{-\lambda \psi(t)} - \phi_h) \nu(t)$, integration by parts gives

$$\int_0^\infty e^{-\rho t} (\delta e^{-\lambda \psi(t)} - \phi_h) \nu(t) dt = \rho \int_0^\infty e^{-\rho t} G(\psi(t)) dt \leq \rho \int_0^\infty e^{-\rho t} G(\min\{t, \bar{T}\}) dt,$$

using $\psi(t) \leq \min\{t, \bar{T}\}$. The bound is attained by the optimal solution $\nu^*(t)$. Q.E.D.

LEMMA D14 (Building blocks). Fix $b > 0$, $k \leq 0$, $S \in \mathbb{R}$, and let $U(w) = S + k(\phi_h - bw)^{1+\lambda_h/b}$ on $w \in [0, \phi_h/b]$. Define

$$Q[U](w) := U(w) - wU'(w), \quad T_a[U](w) := U(w) + \frac{\phi_h - aw}{a + \lambda_h} U'(w) \quad (a > 0).$$

Then:

(i) U is increasing and concave, and with $x := (\phi_h - bw)/\phi_h \in [0, 1]$,

$$T_a[U](w) = S + k \phi_h^{1+\lambda_h/b} \frac{a-b}{b(a+\lambda_h)} (b + \lambda_h - \lambda_h x) x^{\lambda_h/b}.$$

(ii) $Q[U]$ is nondecreasing with $Q[U](0) = U(0)$; $T_a[U]$ is decreasing in w if $a < b$, identically S if $a = b$, and increasing in w if $a > b$.

(iii) For any differentiable V , the left-hand side of (D.1), written $g(w; x_a, x_h)$, satisfies

$$\begin{aligned} g(w; 0, 0) &= -rQ[V](w), & g(w; 1, 0) &= (r + \lambda_a)(S_A - Q[V](w)), \\ g(w; 0, 1) &= (r + \lambda_h)(S_H - T_r[V](w)), & g(w; 1, 1) &= (r + \lambda_a + \lambda_h)(S_{AH} - T_{r+\lambda_a}[V](w)). \end{aligned} \quad (\text{D.4})$$

Proof: (i) With $\beta := 1 + \lambda_h/b > 1$, $U'(w) = -k\beta b(\phi_h - bw)^{\beta-1} \geq 0$ and $U''(w) = k\beta(\beta-1)b^2(\phi_h - bw)^{\beta-2} \leq 0$ since $k \leq 0$. The formula for $T_a[U]$ follows by substituting U and U' and simplifying. (ii) $Q[U]'(w) = -wU''(w) \geq 0$. For T_a : $h_b(x) := (b + \lambda_h - \lambda_h x)x^{\lambda_h/b}$ satisfies $h'_b(x) = \frac{\lambda_h(\lambda_h+b)}{b}x^{\lambda_h/b-1}(1-x) \geq 0$, and x is decreasing in w ; since $k \leq 0$, the sign of $dT_a[U]/dw$ equals the sign of $a-b$ (and $T_a[U] \equiv S$ when $a=b$). (iii) Direct substitution of the definitions of S_A , S_H , S_{AH} into $g(w; x_a, x_h)$ for each (x_a, x_h) . Q.E.D.

D.1. Proof of Lemma A1

When $\nu_h(t) = 1$, Lemma 1 implies $L(t) \geq w(t; \Gamma) + \phi_h/\lambda_h$ for any incentive-compatible contract. Combining with (2),

$$\dot{w}(t; \Gamma) \leq (r + \lambda_a \nu_a(t) + \lambda_h \nu_h(t))w(t; \Gamma) - \lambda_h \nu_h(t) \left(w(t; \Gamma) + \frac{\phi_h}{\lambda_h} \right) = (r + \lambda_a \nu_a(t))w(t; \Gamma) - \phi_h \nu_h(t).$$

Multiplying by $e^{-rt - \lambda_a \psi_a(t)}$ yields

$$\frac{d}{dt} (e^{-rt - \lambda_a \psi_a(t)} w(t; \Gamma)) \leq -\phi_h \nu_h(t) e^{-rt - \lambda_a \psi_a(t)}.$$

For any $T \geq 0$, integrating over $[0, T]$ gives

$$e^{-rT - \lambda_a \psi_a(T)} w(T; \Gamma) - w(0; \Gamma) \leq -\phi_h \int_0^T \nu_h(t) e^{-rt - \lambda_a \psi_a(t)} dt,$$

which yields

$$w(0; \Gamma) \geq \phi_h \int_0^T \nu_h(t) e^{-rt - \lambda_a \psi_a(t)} dt + e^{-rT - \lambda_a \psi_a(T)} w(T; \Gamma) \geq \phi_h \int_0^T \nu_h(t) e^{-rt - \lambda_a \psi_a(t)} dt,$$

where the last inequality holds because of individual rationality (IR). Letting $T \rightarrow \infty$ proves the claim. Q.E.D.

D.2. Proof of Lemma A3

First, the assumptions imply $S_H \geq S_{AH} > S_A$: since $\lambda_a R - c_a \geq 0$ and $S_H > S_A$,

$$(\lambda_h R - c_h)(\lambda_a + r) > (\lambda_a R - c_a)(\lambda_h + r) \geq (\lambda_a R - c_a)\lambda_h;$$

adding $(\lambda_a + r)(\lambda_a R - c_a)$ to both sides and dividing by $(\lambda_a + r)(\lambda_a + \lambda_h + r)$ gives $S_{AH} > S_A$.

Let $g(\eta) := \partial_w f_2(w; \eta)|_{w=\eta} - f'_1(\eta)$ and $h(\eta) := (\phi_h - r\eta)g(\eta)$; since $\phi_h - r\eta > 0$ on the relevant interval, the zeros of g and h coincide. Using $\partial_w f_2(w; \eta)|_{w=\eta} = -\frac{\lambda_h + r}{\phi_h - r\eta}(f_1(\eta) - S_H)$, a direct computation gives

$$h(\eta) = (\lambda_h + r)(S_H - S_{AH}) + \lambda_a(S_A - S_{AH}) \left(1 + \frac{\lambda_h \eta}{\phi_h} \right) \left(1 - \frac{(\lambda_a + r)\eta}{\phi_h} \right)^{\frac{\lambda_h}{\lambda_a + r}}, \quad (\text{D.5})$$

and

$$h'(\eta) = \frac{\lambda_a \lambda_h (\lambda_a + \lambda_h + r)}{\phi_h^2} (S_{AH} - S_A) \eta \left(1 - \frac{(\lambda_a + r)\eta}{\phi_h} \right)^{\frac{\lambda_h}{\lambda_a + r} - 1} > 0 \quad \text{on } \left(0, \frac{\phi_h}{\lambda_a + r} \right), \quad (\text{D.6})$$

since $S_{AH} > S_A$. Moreover $h(0) = -rS_A \leq 0$ and $h(\phi_h/(\lambda_a + r)) = (\lambda_h + r)(S_H - S_{AH}) \geq 0$. Therefore, there exists a unique zero $\eta_0 \in [0, \phi_h/(\lambda_a + r)]$. Q.E.D.

D.3. Proof of Lemma A4

We use the transforms and identities of Lemma D14: write $f_2(w) := f_2(w; \eta_0)$.

Case 1 ($V \equiv S_A$). Here $Q[V] = T_a[V] = S_A$ for all a , so by (D.4): $g(w; 0, 0) = -rS_A \leq 0$, $g(w; 1, 0) = 0$, $g(w; 0, 1) = (r + \lambda_h)(S_H - S_A) \leq 0$, and $g(w; 1, 1) = (r + \lambda_a + \lambda_h)(S_{AH} - S_A) \leq 0$.

Cases 2 and 3. In both cases $S_{AH} > S_A$ (in Case 3, it holds by the preliminary step of the proof of Lemma A3). Each piece of V is a building block of Lemma D14: f_1 with $(b, S, k) = (\lambda_a + r, S_{AH}, (S_A - S_{AH})/\phi_h^2)$ and $k < 0$; f_2 with $(b, S) = (r, S_H)$ and coefficient $k = (f_1(\eta_0) - S_H)/(\phi_h - r\eta_0)^{(\lambda_h + r)/r} \leq 0$, since $f_1 \leq S_{AH} \leq S_H$ in Case 3; and the constant pieces with $k = 0$. Differentiability: at the interior pasting point η_0 by construction; at $w = \phi_h/(\lambda_a + r)$ (Case 2) and $w = \phi_h/r$ (Case 3) the power-piece derivative vanishes, matching the constant piece. Hence V is C^1 , increasing, concave (concave pieces with continuous derivative), and nonnegative with $V(0) = S_A \geq 0$.

Because V is C^1 and piecewise a building block, $Q[V]$ and $T_a[V]$ are continuous, and the monotonicity in Lemma D14(ii) holds piecewise. We verify (D.4) term by term.

$g(w; 0, 0)$ and $g(w; 1, 0)$: $Q[V]$ is globally nondecreasing with $Q[V](0) = V(0) = S_A \geq 0$, so $Q[V](w) \geq S_A$ for all w , giving $g(w; 0, 0) = -rQ[V](w) \leq 0$ and $g(w; 1, 0) = (r + \lambda_a)(S_A - Q[V](w)) \leq 0$.

$g(w; 0, 1)$: we show $T_r[V] \geq S_H$ everywhere. Case 2: on the f_1 -piece ($b = \lambda_a + r > r$), $T_r[V]$ is decreasing with limit $V = S_{AH}$ at $w = \phi_h/(\lambda_a + r)$ (where $V' = 0$), so $T_r[V] \geq S_{AH} \geq S_H$; on the constant piece $T_r[V] = S_{AH} \geq S_H$. Case 3: on $[0, \eta_0]$, $T_r[f_1]$ is decreasing with $T_r[f_1](\eta_0) = T_r[f_2](\eta_0) = S_H$ (since $T_r[f_2] \equiv S_H$ by $a = b = r$), so $T_r[V] \geq S_H$; on the f_2 -piece and on the constant piece $T_r[V] \equiv S_H$.

$g(w; 1, 1)$: we show $T_{r+\lambda_a}[V] \geq S_{AH}$ everywhere. On any f_1 -piece, $a = b = \lambda_a + r$ gives $T_{r+\lambda_a}[f_1] \equiv S_{AH}$, so $g(w; 1, 1) = 0$. In Case 2, on the constant piece $T_{r+\lambda_a}[V] = S_{AH}$, so $g = 0$. In Case 3, on the f_2 -piece ($a = r + \lambda_a > b = r$), $T_{r+\lambda_a}[f_2]$ is increasing from $T_{r+\lambda_a}[f_2](\eta_0) = T_{r+\lambda_a}[f_1](\eta_0) = S_{AH}$, hence $\geq S_{AH}$; on the constant piece $T_{r+\lambda_a}[V] = S_H \geq S_{AH}$.

Thus (D.1) holds in every case and region. Q.E.D.

D.4. Proof of Lemma A5

Region $\lambda_a \geq \hat{\lambda}_a$. On any f_1 -piece,

$$f'_1(0) = \frac{(\lambda_a + \lambda_h + r)(S_{AH} - S_A)}{\phi_h} = \frac{\lambda_h(rR + c_a) - c_h(\lambda_a + r)}{\phi_h(\lambda_a + r)} \leq 1 \quad \text{for } \lambda_a \geq \hat{\lambda}_a,$$

and f'_1 is decreasing, so $f_1(w) - w$ is decreasing; similarly $\partial_w f_2(w; \eta_0)|_{w=\eta_0} = f'_1(\eta_0) \leq f'_1(0) \leq 1$ and $\partial_w f_2$ is decreasing, so $f_2(w; \eta_0) - w$ is decreasing; the constant pieces are trivially decreasing. Hence $w^* = 0$ and $\bar{P} = V(0) = S_A$.

Region $\lambda_a < \hat{\lambda}_a$. Here $S_{AH} - S_A = \frac{\lambda_h(rR + c_a) - c_h(r + \lambda_a)}{(r + \lambda_a + \lambda_h)(r + \lambda_a)} > 0$, so only Cases 2 and 3 arise, and $S_H - S_{AH} = \frac{c_a(r + \lambda_h) - \lambda_a(rR + c_h)}{(r + \lambda_h)(r + \lambda_a + \lambda_h)}$ implies $S_{AH} > \max\{S_H, S_A\}$ for $\lambda_a^* < \lambda_a < \hat{\lambda}_a$, while $S_H \geq S_{AH}$, $S_H > S_A$ for $\lambda_a \leq \lambda_a^*$.

If $\lambda_a^* < \lambda_a < \hat{\lambda}_a$ (Case 2): the maximizer of $V(w) - w$ lies on the f_1 -piece, where f_1 is strictly concave with $f'_1(0) = 1/K > 1$ and $f'_1(\phi_h/(\lambda_a + r)) = 0$, so the FOC $f'_1(w) = 1$ has the unique solution

$$w^* = \frac{\phi_h}{\lambda_a + r} \left(1 - K^{\frac{\lambda_a + r}{\lambda_h}}\right) =: w_1^*. \quad (\text{D.7})$$

Substituting into $f_1(w) - w$ gives

$$\bar{P} = f_1(w_1^*) - w_1^* = S_{AH} - \frac{\phi_h}{\lambda_a + \lambda_h + r} K^{\frac{\lambda_a + r}{\lambda_h}} - \frac{\phi_h}{\lambda_a + r} \left(1 - K^{\frac{\lambda_a + r}{\lambda_h}}\right).$$

If $\lambda_a \leq \lambda_a^*$ (Case 3): the maximizer of $V(w) - w$ lies on the f_1 - or the f_2 -piece, whose respective unique interior maximizers are w_1^* in (D.7) and

$$w_2^* = \eta_0 + \frac{\phi_h - r\eta_0}{r} \left[1 - \left(\frac{\phi_h - r\eta_0}{(\lambda_h + r)(S_H - f_1(\eta_0))}\right)^{\frac{r}{\lambda_h}}\right].$$

Since $V(w) - w$ is concave, $w^* = w_2^*$ iff $w_1^* > \eta_0$ and $w^* = w_1^*$ iff $w_1^* \leq \eta_0$. By lemma D15 below, the sign of $w_1^* - \eta_0$ equals the sign of $\Delta(\lambda_a)$, and Δ changes sign at most once, from positive to negative, on $[c_a/R, \lambda_a^*]$. Hence

$$w^* = w_2^* \text{ for } \frac{c_a}{R} \leq \lambda_a < \check{\lambda}_a, \quad w^* = w_1^* \text{ for } \check{\lambda}_a \leq \lambda_a \leq \lambda_a^*, \quad (\text{D.8})$$

and evaluating $V(w^*) - w^*$ gives the first two lines of (A.25). Combining with (D.7) and the region $\lambda_a \geq \hat{\lambda}_a$ proves (A.24)–(A.25).

LEMMA D15. $\Delta(\lambda_a) > 0$ for $c_a/R \leq \lambda_a < \check{\lambda}_a$ and $\Delta(\lambda_a) \leq 0$ for $\check{\lambda}_a \leq \lambda_a \leq \lambda_a^*$; in particular, Δ changes sign at most once on $[c_a/R, \lambda_a^*]$.

Proof: Let $\ell(\lambda_a) := w_1^*(\lambda_a) - \eta_0(\lambda_a)$, continuous on $[c_a/R, \lambda_a^*]$, and recall from the proof of Lemma A3 that the smooth-pasting point is the unique zero of the strictly increasing map h in (D.5). Using $1 - (\lambda_a + r)w_1^*/\phi_h = K^{(\lambda_a + r)/\lambda_h}$, a direct computation gives the key identity

$$h(w_1^*(\lambda_a)) = \Delta(\lambda_a).$$

Since h is strictly increasing with $h(\eta_0) = 0$, it follows that $\Delta(\lambda_a) > 0 \iff w_1^* > \eta_0 \iff \ell(\lambda_a) > 0$, and $\Delta(\lambda_a) \leq 0 \iff \ell(\lambda_a) \leq 0$.

We show every zero of ℓ is a strict downward crossing. Fix $\bar{\lambda}_a$ with $\ell(\bar{\lambda}_a) = 0$ and set $\bar{w} := w_1^*(\bar{\lambda}_a) = \eta_0(\bar{\lambda}_a)$, $A := \phi_h - (\bar{\lambda}_a + r)\bar{w}$, $B := \phi_h - r\bar{w}$, $z := A/\phi_h$. Differentiating the FOC $f_1'(w_1^*(\lambda_a); \lambda_a) \equiv 1$ and using $f_1''(\bar{w}) = -\lambda_h/A$,

$$w_1^{*'}(\bar{\lambda}_a) = \frac{A}{\lambda_h} \partial_{\lambda_a} f_1'(\bar{w}; \bar{\lambda}_a).$$

Next, define $\Psi(\eta, \lambda_a) := \partial_w f_2(w; \eta; \lambda_a)|_{w=\eta} - f_1'(\eta; \lambda_a)$, so that $\Psi(\eta_0(\lambda_a), \lambda_a) \equiv 0$. At the crossing point, smooth pasting gives $(\lambda_h + r)(S_H - f_1(\bar{w})) = B$, and a direct computation yields

$$\Psi_\eta(\bar{w}, \bar{\lambda}_a) = -\frac{\lambda_h}{B} + \frac{\lambda_h}{A} = \frac{\lambda_h \bar{\lambda}_a \bar{w}}{AB} > 0, \quad \Psi_{\lambda_a}(\bar{w}, \bar{\lambda}_a) = -\partial_{\lambda_a} f_1'(\bar{w}) - \frac{(\lambda_h + r)\partial_{\lambda_a} f_1(\bar{w})}{B}.$$

The implicit function theorem then gives

$$\eta_0'(\bar{\lambda}_a) = \frac{A}{\lambda_h \bar{\lambda}_a \bar{w}} \left(B \partial_{\lambda_a} f_1'(\bar{w}) + (\lambda_h + r) \partial_{\lambda_a} f_1(\bar{w}) \right),$$

and hence, using $\bar{\lambda}_a \bar{w} - B = -A$,

$$\ell'(\bar{\lambda}_a) = -\frac{A}{\lambda_h \bar{\lambda}_a \bar{w}} \left(A \partial_{\lambda_a} f_1'(\bar{w}) + (\lambda_h + r) \partial_{\lambda_a} f_1(\bar{w}) \right).$$

Substituting the explicit expressions for $\partial_{\lambda_a} f_1'$ and $\partial_{\lambda_a} f_1$ and simplifying, using $A(\bar{\lambda}_a + r)/M = z^\beta$ with $M := \lambda_h(rR + c_a) - c_h(\bar{\lambda}_a + r) > 0$ (valid since $\lambda_a \leq \lambda_a^* < \bar{\lambda}_a$) and $\beta := \frac{\lambda_a + \lambda_h + r}{\lambda_a + r} > 1$, one obtains

$$A \partial_{\lambda_a} f_1'(\bar{w}) + (\lambda_h + r) \partial_{\lambda_a} f_1(\bar{w}) = \frac{(\lambda_h + r)(rR + c_a + c_h)}{(\bar{\lambda}_a + \lambda_h + r)^2} (1 - z^\beta) + \frac{r\bar{w}}{\bar{\lambda}_a + r} + \frac{r(rR + c_a)A}{(\bar{\lambda}_a + r)M} + \frac{-A\lambda_h \bar{\lambda}_a \ln z}{(\bar{\lambda}_a + r)^2(\bar{\lambda}_a + \lambda_h + r)},$$

in which every term is strictly positive. Hence $\ell'(\bar{\lambda}_a) < 0$, so ℓ (equivalently Δ) crosses zero at most once and only downward. The claim then follows from the definition of $\bar{\lambda}_a$ and continuity of Δ . Q.E.D.

D.5. Proof of Lemma A6

Denote the candidate payoff-to-go in the lemma by $\hat{w}(t)$, so that $L(t) = \hat{w}(t) + \phi_h/\lambda_h$ for $t \leq T_2$. Under $\Gamma_{HA}(T_1, T_2)$ the effort processes are

$$(\nu_h(t), \nu_a(t)) = \begin{cases} (1, 0), & t \in [0, T_1], \\ (1, 1), & t \in (T_1, T_2], \\ (0, 1), & t \in (T_2, \infty). \end{cases} \quad (\text{D.9})$$

One verifies directly that $\hat{w}$ satisfies (2) with $\hat{w}(\infty) = 0$; hence Lemma D12 gives $w(t) := w(t; \Gamma_{HA}(T_1, T_2)) = \hat{w}(t)$. Since $L(t) - w(t) = \phi_h/\lambda_h$, (IC(t)) holds with equality for all t , so the contract is incentive compatible by Lemma 1.

For the principal's payoff, apply the societal-value decomposition (D.3) under (D.9) and evaluate the three integrals:

$$\begin{aligned} V &= \frac{\lambda_h R - c_h}{r + \lambda_h} \left(1 - e^{-(r+\lambda_h)T_1} \right) + \frac{(\lambda_h + \lambda_a)R - c_h - c_a}{r + \lambda_h + \lambda_a} e^{-(r+\lambda_h)T_1} \left(1 - e^{-(r+\lambda_h+\lambda_a)(T_2-T_1)} \right) \\ &\quad + \frac{\lambda_a R - c_a}{r + \lambda_a} e^{-(r+\lambda_h)T_1 - (r+\lambda_h+\lambda_a)(T_2-T_1)} \\ &= S_H (1 - e^{-(\lambda_h+r)T_1}) + e^{-(\lambda_h+r)T_1} \left[S_{AH} + (S_A - S_{AH}) e^{-(\lambda_a+\lambda_h+r)(T_2-T_1)} \right], \end{aligned}$$

and $P(\Gamma_{HA}(T_1, T_2)) = V - w(0)$. Q.E.D.

D.6. Proof of Lemma B7

The individual bound (B.4) is proved exactly as Lemma A1 and hence omitted. Summing (B.4) over $i = 1, 2$,

$$w_1(0; \Gamma) + w_2(0; \Gamma) \geq \phi_h \int_0^\infty e^{-rt - \lambda_h H(t)} \left[e^{\lambda_h \psi_1(t)} d\psi_1(t) + e^{\lambda_h \psi_2(t)} d\psi_2(t) \right] = \frac{\phi_h}{\lambda_h} \int_0^\infty e^{-rt - \lambda_h H(t)} d\pi(t),$$

where $\pi(t) := e^{\lambda_h \psi_1(t)} + e^{\lambda_h \psi_2(t)}$. By convexity of the exponential and $\psi_1 + \psi_2 = H$, $\pi(t) \geq G(t) := 2e^{\lambda_h H(t)/2}$, with $\pi(0) = G(0) = 2$. Since $q(t) := e^{-rt - \lambda_h H(t)}$ is nonincreasing, integration by parts on $[0, T]$ gives

$$\int_0^T q(t) d(\pi - G)(t) = q(T)(\pi(T) - G(T)) - \int_0^T (\pi(t) - G(t)) dq(t) \geq 0,$$

so, letting $T \rightarrow \infty$, $\int_0^\infty q(t) d\pi(t) \geq \int_0^\infty q(t) dG(t)$. Since $dG(t) = \lambda_h e^{\lambda_h H(t)/2} dH(t)$, the aggregate bound (B.5) follows. Q.E.D.

D.7. Proof of Lemma B8

For $w \in [0, \frac{\phi_h}{\lambda_a + r}]$,

$$\frac{\partial f_1(w; \lambda_a)}{\partial w} = C(\lambda_a)B(\lambda_a), \quad B(\lambda_a) := \left(1 - \frac{(\lambda_a + r)w}{\phi_h}\right)^{\lambda_h/(\lambda_a + r)}, \quad C(\lambda_a) := \frac{\lambda_h(rR + c_a) - c_h(\lambda_a + r)}{\phi_h(\lambda_a + r)}.$$

Both factors are positive (since $S_{AH} > S_A$) and decreasing in λ_a :

$$C'(\lambda_a) = \frac{-\lambda_h(rR + c_a)}{\phi_h(\lambda_a + r)^2} < 0, \quad \frac{B'(\lambda_a)}{B(\lambda_a)} = -\frac{\lambda_h}{(\lambda_a + r)^2} \left(\ln u + \frac{1}{u} - 1\right) < 0, \quad u := 1 - \frac{(\lambda_a + r)w}{\phi_h} \in (0, 1),$$

using that $g(u) := \ln u + 1/u - 1$ is strictly decreasing on $(0, 1)$ with $g(1) = 0$, hence $g(u) > 0$. Thus $\partial f_1/\partial w$ is decreasing in λ_a . For c_a :

$$\frac{\partial f_1(w; c_a)}{\partial w} = \frac{\lambda_a + \lambda_h + r}{\phi_h - (\lambda_a + r)w} (S_{AH}(c_a) - f_1(w; c_a)) = \frac{\lambda_a + \lambda_h + r}{\phi_h - (\lambda_a + r)w} (S_{AH}(c_a) - S_A(c_a)) \left(1 - \frac{(\lambda_a + r)w}{\phi_h}\right)^{\frac{\lambda_a + \lambda_h + r}{\lambda_a + r}},$$

which is increasing in c_a , since $S_{AH}(c_a) - S_A(c_a) = \frac{\lambda_h r R + \lambda_h c_a - c_h(\lambda_a + r)}{(\lambda_a + \lambda_h + r)(\lambda_a + r)}$ is increasing in c_a .

Next, $f_1(w; \lambda_a)$ is increasing in λ_a . From the ODE $(\phi_h - (\lambda_a + r)w) \partial_w f_1 = (\lambda_a + \lambda_h + r)(S_{AH} - f_1)$, differentiating in λ_a gives

$$(\phi_h - (\lambda_a + r)w) \partial_{w\lambda_a}^2 f_1 + (\lambda_a + \lambda_h + r) \partial_{\lambda_a} f_1 = w \partial_w f_1 + S_{AH} - f_1 + (\lambda_a + \lambda_h + r) S'_{AH}(\lambda_a) > 0,$$

where the inequality holds because $f_1(w; \lambda_a)$ is increasing in w . And the initial condition is $\frac{\partial f_1(w; \lambda_a)}{\partial \lambda_a}|_{w=0} = S'_{AI}(\lambda_a) \geq 0$. If $\partial_{\lambda_a} f_1$ were negative somewhere on $(0, \phi_h/(\lambda_a + r)]$, continuity would give a first zero w_0 , at which the displayed equation forces $\partial_{w\lambda_a}^2 f_1|_{w=w_0} > 0$, contradicting with the property that $\frac{\partial f_1(w; \lambda_a)}{\partial w}$ is decreasing in λ_a we just established.

For f_2 on $[\eta_0, \phi_h/r]$ with $S_H \geq S_{AH}$, $S_H > S_A$: write

$$f_2(w; \lambda_a) = S_H + K(\lambda_a)(\phi_h - rw)^{(\lambda_h + r)/r}, \quad K(\lambda_a) := \frac{f_1(\eta_0(\lambda_a); \lambda_a) - S_H}{(\phi_h - r\eta_0(\lambda_a))^{(\lambda_h + r)/r}},$$

so f_2 depends on λ_a only through K . By taking derivative, we have

$$K'(\lambda_a) = \frac{1}{(\phi_h - r\eta_0(\lambda_a))^{(\lambda_h + r)/r}} \left[\frac{\partial f_1(w; \lambda_a)}{\partial \lambda_a} \Big|_{w=\eta_0(\lambda_a)} + \eta'_0(\lambda_a) \left(\frac{\partial f_1(w; \lambda_a)}{\partial w} \Big|_{w=\eta_0(\lambda_a)} + \frac{(\lambda_h + r)(f_1(\eta_0(\lambda_a); \lambda_a) - S_H)}{\phi_h - r\eta_0(\lambda_a)} \right) \right]$$

Using the smooth-pasting identity $\partial_w f_2|_{w=\eta_0} = -\frac{(\lambda_h + r)(f_1(\eta_0) - S_H)}{\phi_h - r\eta_0} = f'_1(\eta_0)$, the terms multiplying $\eta'_0(\lambda_a)$ cancel, leaving

$$K'(\lambda_a) = \frac{\partial_{\lambda_a} f_1(w; \lambda_a)|_{w=\eta_0(\lambda_a)}}{(\phi_h - r\eta_0(\lambda_a))^{(\lambda_h + r)/r}} \geq 0.$$

Hence $\partial_{\lambda_a} f_2 = K'(\lambda_a)(\phi_h - rw)^{(\lambda_h + r)/r} \geq 0$ and $\partial_{w\lambda_a}^2 f_2 = -(\lambda_h + r)K'(\lambda_a)(\phi_h - rw)^{\lambda_h/r} \leq 0$. Lastly,

$$f_2(w; c_a) = S_H + (f_1(\eta_0(c_a); c_a) - S_H) \left(1 - \frac{r(w - \eta_0(c_a))}{\phi_h - r\eta_0(c_a)}\right)^{\frac{\lambda_h + r}{r}} = S_H + \frac{(f_1(\eta_0(c_a); c_a) - S_H)}{(\phi_h - r\eta_0(c_a))^{(\lambda_h + r)/r}} (\phi_h - rw)^{(\lambda_h + r)/r}.$$

Therefore, $f_2(w; c_a)$ depends on c_a only through $\hat{K}(c_a) := \frac{(f_1(\eta_0(c_a); c_a) - S_H)}{(\phi_h - r\eta_0(c_a))^{(\lambda_h + r)/r}}$. By taking derivative, and using the same cancellation argument in c_a , we have

$$\hat{K}'(c_a) = \frac{1}{(\phi_h - r\eta_0(c_a))^{(\lambda_h + r)/r}} \frac{\partial f_1(w; c_a)}{\partial c_a} \Big|_{w=\eta_0(c_a)} < 0$$

where the inequality holds because

$$\frac{\partial f_1(w; c_a)}{\partial c_a} = -\frac{1}{\lambda_a + \lambda_h + r} + \left(-\frac{1}{\lambda_a + r} + \frac{1}{\lambda_a + \lambda_h + r}\right) \left(1 - \frac{(\lambda_a + r)w}{\phi_h}\right)^{\frac{\lambda_a + \lambda_h + r}{\lambda_a + r}} < 0,$$

and hence $\partial_{wc_a}^2 f_2 = -(\lambda_h + r)\hat{K}'(c_a)(\phi_h - rw)^{\lambda_h/r} \geq 0$. Q.E.D.

D.8. Proof of Lemma B9

By definition of $\lambda_a^A(c_a)$,

$$w^*(\Gamma_{HA}^*(\lambda_a^A(c_a)), c_a) = w_{HH}^* \in \arg \max_{w \geq 0} \{V(w; \lambda_a^A(c_a), c_a) - w\}. \quad (D.10)$$

f₂-branch case. If the maximizer lies on f_2 , i.e., $\arg \max_{w \geq 0} \{V(w; \lambda_a^A(c_a), c_a) - w\} = \arg \max_{w \geq 0} \{f_2(w; \lambda_a^A(c_a), c_a) - w\}$ the FOC $\frac{\lambda_h + r}{\phi_h - r w_{HH}^*} (S_H - f_2(w_{HH}^*)) = 1$ gives $f_2(w_{HH}^*) = S_H - \frac{\phi_h - r w_{HH}^*}{\lambda_h + r}$, so

$$Q(c_a) = f_2(w_{HH}^*) - w_{HH}^* = \frac{\lambda_h R - c_h - \phi_h - \lambda_h w_{HH}^*}{\lambda_h + r},$$

and hence

$$Q(c_a) - P_{HH}^* = \frac{\lambda_h R - c_h - \phi_h - \lambda_h w_{HH}^*}{\lambda_h + r} - \frac{2(\lambda_h R - c_h - \phi_h - \lambda_h w_{HH}^*)}{2\lambda_h + r} = -\frac{r(\lambda_h R - c_h - \phi_h - \lambda_h w_{HH}^*)}{(\lambda_h + r)(2\lambda_h + r)} < 0.$$

f₁-branch case. If the maximizer lies on f_1 , the FOC

$$\frac{\lambda_a^A + \lambda_h + r}{\phi_h} (S_{AH} - S_A) \left(1 - \frac{(\lambda_a^A + r)w_{HH}^*}{\phi_h}\right)^{\frac{\lambda_h}{\lambda_a^A + r}} = 1 \quad (D.11)$$

(with all objects evaluated at $(\lambda_a^A(c_a), c_a)$) yields

$$Q(c_a) = \frac{(\lambda_a^A + \lambda_h)R - (c_a + c_h) - \phi_h - \lambda_h w_{HH}^*}{\lambda_a^A + \lambda_h + r}, \quad (D.12)$$

and, solving (D.11) for c_a ,

$$c_a = \frac{\lambda_a^A + r}{\lambda_h} \left(c_h + \phi_h \left(1 - \frac{(\lambda_a^A + r)w_{HH}^*}{\phi_h}\right)^{-\frac{\lambda_h}{\lambda_a^A + r}} \right) - rR.$$

Substituting into (D.12), $Q(c_a)$ depends on c_a only through $x := \lambda_a^A(c_a) + r$, via

$$\hat{Q}(x) = R - \frac{\frac{x}{\lambda_h} (c_h + \phi_h F(x)) + c_h + \phi_h + \lambda_h w_{HH}^*}{x + \lambda_h}, \quad F(x) := q(x)^{-\lambda_h/x}, \quad q(x) := 1 - \frac{w_{HH}^* x}{\phi_h} \in (0, 1].$$

It therefore suffices to show that $\hat{Q}$ is decreasing in x and $\lambda_a^A(c_a)$ is increasing in c_a .

Differentiating and simplifying,

$$\hat{Q}'(x) = \frac{\Psi(x)}{(x + \lambda_h)^2}, \quad \Psi(x) := \phi_h + \lambda_h w_{HH}^* - \phi_h F(x) \left[1 + \left(1 + \frac{\lambda_h}{x}\right) \left(\ln q(x) + \frac{1 - q(x)}{q(x)} \right) \right].$$

For $q \in (0, 1]$, $\ln q \geq (q - 1)/q$ gives $\ln q + \frac{1 - q}{q} \geq 0$, so the bracket is at least 1 and $\Psi(x) \leq \phi_h + \lambda_h w_{HH}^* - \phi_h F(x)$. Also $\ln q \leq q - 1$ gives $-\ln q(x) \geq 1 - q(x) = \frac{w_{HH}^* x}{\phi_h}$, so $F(x) \geq e^{\lambda_h w_{HH}^* / \phi_h}$, whence, using $e^t \geq 1 + t$,

$$\Psi(x) \leq \phi_h + \lambda_h w_{HH}^* - \phi_h e^{\lambda_h w_{HH}^* / \phi_h} \leq 0,$$

i.e., $\hat{Q}'(x) \leq 0$. Finally, since $w^*(0; \Gamma_{HA}^*(\lambda_a, c_a))$ is increasing in c_a and decreasing in λ_a (proof of Proposition 4), the defining condition in (4) implies $\lambda_a^A(c_a)$ is increasing in c_a . Q.E.D.

D.9. Proof of Lemma C11

Let $z(t) := e^{-rt - \lambda_a \psi_a(t)}$. Problem (P_a) becomes the deterministic control program

$$\begin{aligned} \max \quad & \tilde{S}_A + \int_0^{T_3^*} z(t) \left[\delta e^{-\lambda_h t} - \phi_h - e^{-\lambda_h t} (r \tilde{S}_A x^N(t) + \eta x^{AC}(t)) \right] dt, \\ \text{s.t.} \quad & \dot{z}(t) = -(r + \lambda_a q(t)) z(t), \quad q(t) := x^A(t) + p x^{AC}(t), \quad z(0) = 1, \\ & x^N + x^{AC} + x^A = 1, \quad x^N, x^{AC}, x^A \geq 0. \end{aligned} \quad (D.13)$$

Step 1 (Maximum principle). With costate μ for z , the Hamiltonian is

$$\mathcal{H} = z \left[\delta e^{-\lambda_h t} - \phi_h - \mu r - e^{-\lambda_h t} \left(r \tilde{S}_A x^N + (\eta + \lambda_a p e^{\lambda_h t} \mu) x^{AC} + \lambda_a e^{\lambda_h t} \mu x^A \right) \right].$$

Let $\beta(t) := e^{\lambda_h t} \mu(t)$. Since the controls enter linearly and $z > 0$, the maximizer selects the mode with the smallest cost among $r\tilde{S}_A$ (mode N), $\eta + \lambda_a p \beta(t)$ (mode AC), and $\lambda_a \beta(t)$ (mode A); Define

$$\beta_1 := \frac{\eta}{\lambda_a(1-p)}, \quad \beta_2 := \frac{r\tilde{S}_A - \eta}{\lambda_a p}, \quad \beta_3 := \frac{r\tilde{S}_A}{\lambda_a}.$$

then,

$$x^{N*}(t) = 1 \iff \beta(t) \geq \beta_2 \vee \beta_3, \quad x^{AC*}(t) = 1 \iff \beta_1 \leq \beta(t) \leq \beta_2, \quad x^{A*}(t) = 1 \iff \beta(t) \leq \beta_3 \wedge \beta_1.$$

Step 2 (Costate monotonicity). The costate equation $\dot{\mu} = -\partial \mathcal{H} / \partial z = e^{-\lambda_h t} \min\{r\tilde{S}_A, \eta + \lambda_a p \beta(t), \lambda_a \beta(t)\} + \phi_h + r\mu - \delta e^{-\lambda_h t}$ with transversality $\mu(\bar{T}_h) = 0$ gives

$$\dot{\beta}(t) = \lambda_h e^{\lambda_h t} \mu(t) + e^{\lambda_h t} \dot{\mu}(t) = \underbrace{(\lambda_h + r)\beta(t) + \min\{r\tilde{S}_A, \eta + \lambda_a p \beta(t), \lambda_a \beta(t)\} + \phi_h e^{\lambda_h t} - \delta}_{=: F(t, \beta(t))}, \quad \beta(\bar{T}_h) = 0. \quad (\text{D.14})$$

Setting $\pi(t) := \dot{\beta}(t)$, since F is increasing in t and Lipschitz and increasing in β , so almost everywhere $\dot{\pi} = \partial_\beta F \cdot \pi + \partial_t F$ with $\pi(\bar{T}_h) = F(\bar{T}_h, 0) = \min\{r\tilde{S}_A, \eta, 0\} + \phi_h e^{\lambda_h \bar{T}_h} - \delta = 0$. Therefore,

$$\pi(t) = - \int_t^{\bar{T}_h} \exp\left(\int_t^s \partial_\beta F \, dx\right) \partial_\tau F \, d\tau < 0 \quad \text{for } t < \bar{T}_h,$$

i.e., $\beta(t)$ is strictly decreasing in t . As $\beta(t)$ decreases, the pointwise minimization rule changes mode at most twice, in the order $N \rightarrow AC \rightarrow A$ (some intervals possibly empty), yielding switch times $0 \leq T_1^* \leq T_2^* \leq T_3^*$ with the stated structure; combined with $x^{A*}(t) = 1$ for $t \geq T_3^*$, mode A persists on $[T_2^*, \infty)$.

Step 3 (Sufficiency). The maximized Hamiltonian $\tilde{\mathcal{H}}(t, z, \mu) = z \max_x A(t, x, \mu)$ is linear, hence concave, in $z \geq 0$. By the Arrow sufficiency theorem (Kamien and Schwartz 1971), the candidate policy, which maximizes $\mathcal{H}$ pointwise with the costate satisfying (D.14), is globally optimal. Q.E.D.